Approximation to PDE by continuous trial functions

- Defining the PDE problem
- Introduction to the PDE approximation
- Approximation by weighted residual
- Simultaneous approximation to the solution of the PDE in the domain and on its boundary.
- Natural boundary conditions
- Boundary solution methods
- Systems of PDE
- Nonlinear problems

Defining the PDE problem

$$A(\phi) = 0 \qquad \text{in } \Omega$$

L : linear differential operator

$$p \neq p(\phi)$$

$$\mathsf{L}(\phi) = \frac{\partial}{\partial x}\left(\kappa \frac{\partial \phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\kappa \frac{\partial \phi}{\partial y}\right)$$

$$p = Q$$

$$\kappa \neq \kappa(\phi)$$

$$B(\phi) = \mathsf{M}(\phi) + \mathrm{r} = 0 \qquad \text{in } \Gamma$$

$$\begin{cases} \mathsf{M}(\phi) = \phi \quad ; \quad r = -\bar{\phi} & \text{on } \Gamma_\phi \quad \text{DIRICHLET} \\ \mathsf{M}(\phi) = -\kappa \dfrac{\partial \phi}{\partial \eta} \quad ; \quad r = -\bar{q} & \text{on } \Gamma_q \quad \text{NEUMANN} \end{cases}$$

Introduction to the PDE approximation
How to choose the trial functions

$$\phi(x) \cong \hat{\phi}(x) = \psi(x) + \sum_m a_m N_m(x)$$

$$\text{on } \Gamma \quad \begin{cases} \mathsf{M}(\psi) = -r \\ \mathsf{M}(N_m) = 0 \quad ; \quad m = 1,2,\cdots \end{cases}$$

$$\hat{\phi}(x) = \phi(x) \quad \text{for all } x \in \Gamma$$

Introduction to the PDE approximation

The trial functions and its derivatives

$$\phi(x) \cong \hat{\phi}(x) = \psi(x) + \sum_m a_m N_m(x)$$

$$\mathsf{L}\hat{\phi} = \mathsf{L}\psi + \sum_m a_m \mathsf{L} N_m$$

$$\frac{\partial \phi}{\partial x} \cong \frac{\partial \hat{\phi}}{\partial x} = \frac{\partial \psi}{\partial x} + \sum_m a_m \frac{\partial N_m}{\partial x}$$

$$\frac{\partial^2 \phi}{\partial x} \cong \frac{\partial^2 \hat{\phi}}{\partial x} = \frac{\partial^2 \psi}{\partial x} + \sum_m a_m \frac{\partial^2 N_m}{\partial x}$$

N_m has the regularity imposed by L

Approximation by weighted residual

$$R_\Omega = A(\phi) = \mathsf{L}\hat{\phi} + p = \mathsf{L}\psi + \sum_m a_m \mathsf{L}N_m + p$$

$$\int_\Omega W_l \, R_\Omega \, d\Omega = \int_\Omega W_l \left(\mathsf{L}\psi + \sum_m a_m \mathsf{L}N_m + p \right) d\Omega = 0$$

$$\int_\Omega W_l \left(\sum_m a_m \mathsf{L}N_m \right) d\Omega = - \int_\Omega W_l \left(\mathsf{L}\psi + p \right) d\Omega$$

$$K_{lm} a_m = f_l$$

$$K_{lm} = \int_\Omega W_l \mathsf{L}N_m \, d\Omega$$

$$f_l = - \int_\Omega W_l \left(\mathsf{L}\psi + p \right) d\Omega \qquad l, m = 1, 2, \cdots, M$$

Point collocation

Weight function $W_l = \delta(x - x_l)$

$$\int_\Omega W_l R_\Omega d\Omega = \int_\Omega \delta(x - x_l)\left(L\hat{\phi} + p\right) d\Omega =$$

$$= \int_\Omega \delta(x - x_l)\left(\left(L\psi + \sum_m a_m L N_m\right) + p\right) d\Omega$$

$$\therefore \quad \int_\Omega \delta(x - x_l)\left(\sum_m a_m L N_m\right) d\Omega = -\int_\Omega \delta(x - x_l)(L\psi + p)\, d\Omega$$

$$K_{lm}\, a_m = f_l$$

$$K_{lm} = \int_\Omega \delta(x - x_l)\, L N_m\, d\Omega = L N_m(x = x_l)$$

$$f_l = -\int_\Omega \delta(x - x_l)(L\psi + p)\, d\Omega = -\left(L\psi(x = x_l) + p(x = x_l)\right)$$

Subdomain collocation

Weight function $W_l = \chi_{\Omega_l} = \begin{cases} 1 & x \in \Omega_l \\ 0 & x \notin \Omega_l \end{cases}$

$$\int_\Omega W_l R_\Omega d\Omega = \int_\Omega \chi_{\Omega_l} \left(\mathsf{L}\hat{\phi} + p \right) d\Omega =$$

$$= \int_\Omega \chi_{\Omega_l} \left(\left(\mathsf{L}\psi + \sum_m a_m \mathsf{L} N_m \right) + p \right) d\Omega$$

$$\therefore \quad \int_\Omega \chi_{\Omega_l} \left(\sum_m a_m \mathsf{L} N_m \right) d\Omega = - \int_\Omega \chi_{\Omega_l} \left(\mathsf{L}\psi + p \right) d\Omega$$

$$K_{lm}\, a_m = f_l$$

$$K_{lm} = \int_\Omega \chi_{\Omega_l}\, \mathsf{L} N_m \, d\Omega = \int_{\Omega_l} \mathsf{L} N_m d\Omega$$

$$f_l = - \int_\Omega \chi_{\Omega_l} \left(\mathsf{L}\psi + p \right) d\Omega = - \int_{\Omega_l} \left(\mathsf{L}\psi + p \right) d\Omega$$

Galerkin

Weight function $W_l = N_l$

$$\int_\Omega W_l R_\Omega d\Omega = \int_\Omega N_l \left(\mathsf{L}\hat{\phi} + p \right) d\Omega =$$

$$= \int_\Omega N_l \left(\left(\mathsf{L}\psi + \sum_m a_m \mathsf{L} N_m \right) + p \right) d\Omega$$

$$\therefore \quad \int_\Omega N_l \left(\sum_m a_m \mathsf{L} N_m \right) d\Omega = - \int_\Omega N_l \left(\mathsf{L}\psi + p \right) d\Omega$$

$$K_{lm}\, a_m = f_l$$

$$K_{lm} = \int_\Omega N_l \mathsf{L} N_m \, d\Omega$$

$$f_l = - \int_\Omega N_l \left(\mathsf{L}\psi + p \right) d\Omega$$

Example 1- Definition

Find $\hat{\phi}(x)$ solution of the following ODE

$$\frac{d^2\phi}{dx^2} - \phi = 0 \qquad \text{in } \Omega : \{ x; 0 \le x \le 1 \}$$

$$\phi(x = 0) = 0$$

$$\phi(x = 1) = 1 \qquad \text{in } \Gamma : \{ x = 0, x = 1 \}$$

$$L(\phi) = \frac{d^2\phi}{dx^2} - \phi$$

$$p = 0$$

$$M(\phi) + r = 0 \qquad \text{in } \Gamma$$

$$\begin{cases} M(\phi) = \phi & ; \quad r = 0 \quad \text{on } x = 0 \quad \text{DIRICHLET} \\ M(\phi) = \phi & ; \quad r = -1 \quad \text{on } x = 1 \quad \text{DIRICHLET} \end{cases}$$

Example 1- Choosing the approximation

$$\phi(x) \cong \hat{\phi}(x) = \psi(x) + \sum_m a_m N_m(x)$$

$$\psi(x) = x$$

$$N_m(x) = \sin(m\pi\, x)$$

$$\psi(x=0) = \phi(x=0) = 0$$

$$N_m(x=0) = 0$$

$$\psi(x=1) = \phi(x=1) = 1$$

$$N_m(x=1) = 0$$

$$L\hat{\phi} = L\psi + \sum_m a_m L N_m$$

$$L\psi = \frac{d^2\psi}{dx^2} - \psi = 0 - x = -x$$

$$L N_m = \frac{d^2 N_m}{dx^2} - N_m = (m\pi)^2(-\sin(m\pi\, x)) - \sin(m\pi\, x)$$

Example 1- Weighted residual using collocation

$$K_{lm}\, a_m = f_l$$

$$K_{lm} = L N_m (x = x_l)$$

$$L N_m = \sin(m\pi\, x)\left(-1 - (m\pi)^2\right)$$

$$K_{lm} = \sin(m\pi\, x_l)\left(-1 - (m\pi)^2\right)$$

$$L\psi = -x \qquad ; \qquad p = 0$$

$$f_l = -\left(-x_l + 0\right) = x_l$$

$$\therefore \qquad a_m = \left(K^{-1} f\right)_m$$

Example 1- Weighted residual using collocation (see routine Ej_2_1.m)

$$K_{lm}\, a_m = f_l$$

$$K_{lm} = -\sin(m\pi\, x_l)\left(1+(m\pi)^2\right) \quad ; \quad f_l = x_l$$

$$\therefore K = -\begin{vmatrix} \sin(\pi\, x_1)\left(1+\pi^2\right) & \sin(2\pi\, x_1)\left(1+4\pi^2\right) & \cdots & \sin(M\pi\, x_1)\left(1+M^2\pi^2\right) \\ \sin(\pi\, x_2)\left(1+\pi^2\right) & \sin(2\pi\, x_2)\left(1+4\pi^2\right) & \cdots & \sin(M\pi\, x_2)\left(1+M^2\pi^2\right) \\ \vdots & \vdots & \ddots & \vdots \\ \sin(\pi\, x_M)\left(1+\pi^2\right) & \sin(2\pi\, x_M)\left(1+4\pi^2\right) & & \sin(M\pi\, x_M)\left(1+M^2\pi^2\right) \end{vmatrix}$$

$$f = \begin{vmatrix} x_1 & x_2 & \cdots & x_M \end{vmatrix}^T$$

Example 1- Weighted residual using Galerkin (see routine Ej_2_1.m)

$$K_{lm}\, a_m = f_l$$

$$K_{lm} = -\int_0^1 \sin(l\pi x)\sin(m\pi x)\left(1+(m\pi)^2\right)dx \quad ; \quad f_l = \int_0^1 x\sin(l\pi x)dx$$

$$\therefore K = -\begin{vmatrix} \int_0^1 \sin(\pi x)\sin(\pi x)\left(1+\pi^2\right)dx & \int_0^1 \sin(\pi x)\sin(2\pi x)\left(1+4\pi^2\right)dx & \cdots & \int_0^1 \sin(\pi x)\sin(M\pi x)\left(1+M^2\pi^2\right)dx \\ \int_0^1 \sin(2\pi x)\sin(\pi x)\left(1+\pi^2\right)dx & \int_0^1 \sin(2\pi x)\sin(2\pi x)\left(1+4\pi^2\right)dx & \cdots & \int_0^1 \sin(2\pi x)\sin(M\pi x)\left(1+M^2\pi^2\right)dx \\ \vdots & \vdots & \ddots & \vdots \\ \int_0^1 \sin(M\pi x)\sin(\pi x)\left(1+\pi^2\right)dx & \int_0^1 \sin(M\pi x)\sin(2\pi x)\left(1+4\pi^2\right)dx & & \int_0^1 \sin(M\pi x)\sin(M\pi x)\left(1+M^2\pi^2\right)dx \end{vmatrix}$$

$$f = \left| \int_0^1 x\sin(\pi x)dx \quad \int_0^1 x\sin(2\pi x)dx \quad \cdots \quad \int_0^1 x\sin(M\pi x)dx \right|^T$$

since $\int_0^1 \sin(l\pi x)\sin(m\pi x)dx = \frac{1}{2}\delta_{lm}$ by orthogonality

and $\int_0^1 x\sin(l\pi x)\,dx = (-1)^{l+1}\frac{1}{l\pi}$

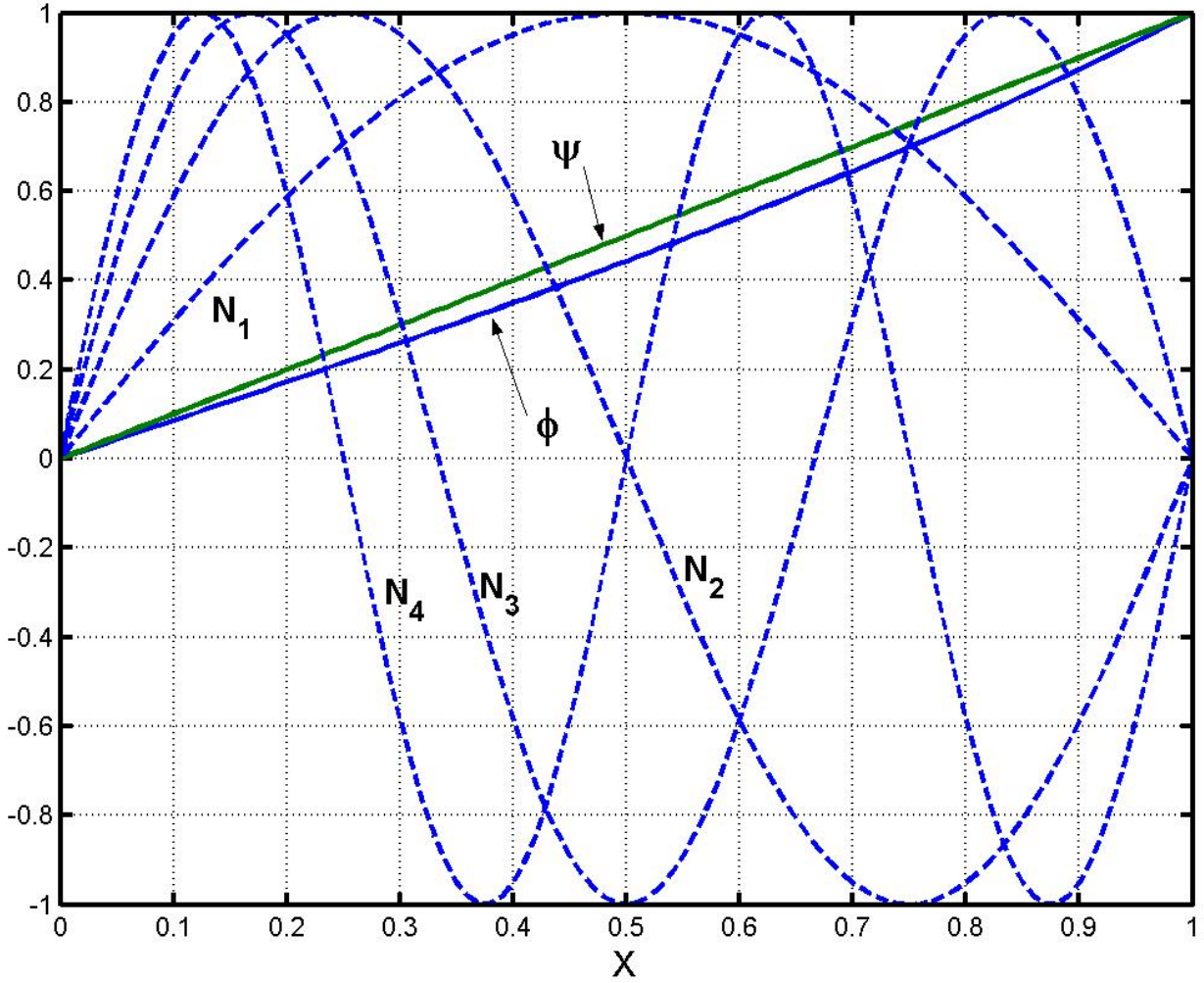
$$\therefore \quad \frac{1}{2}\delta_{lm}\left(1+m^2\pi^2\right)a_m = (-1)^m\frac{1}{m\pi} \quad \Rightarrow \quad a_m = (-1)^m\frac{2}{m\pi\left(1+m^2\pi^2\right)}$$

Example 1- Exact solution and approximation basis

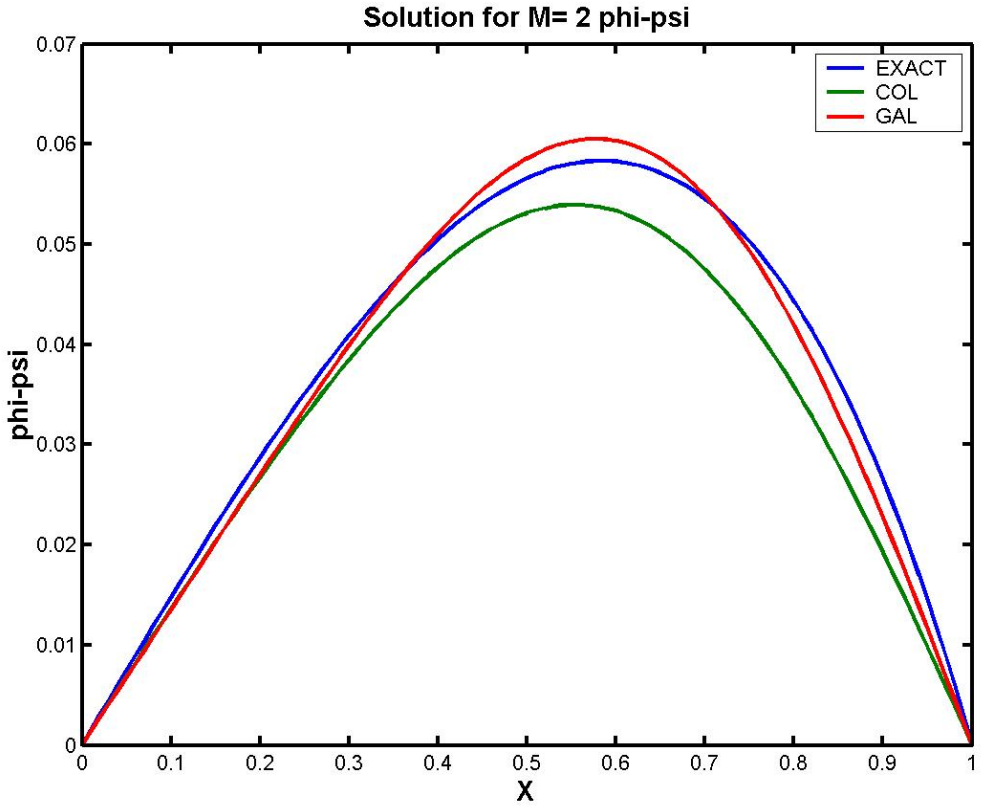
$$\phi^{\text{exact}} = \frac{e^{x} - e^{-x}}{e^{1} - e^{-1}}$$

$$N_m = \sin(m\pi x)$$

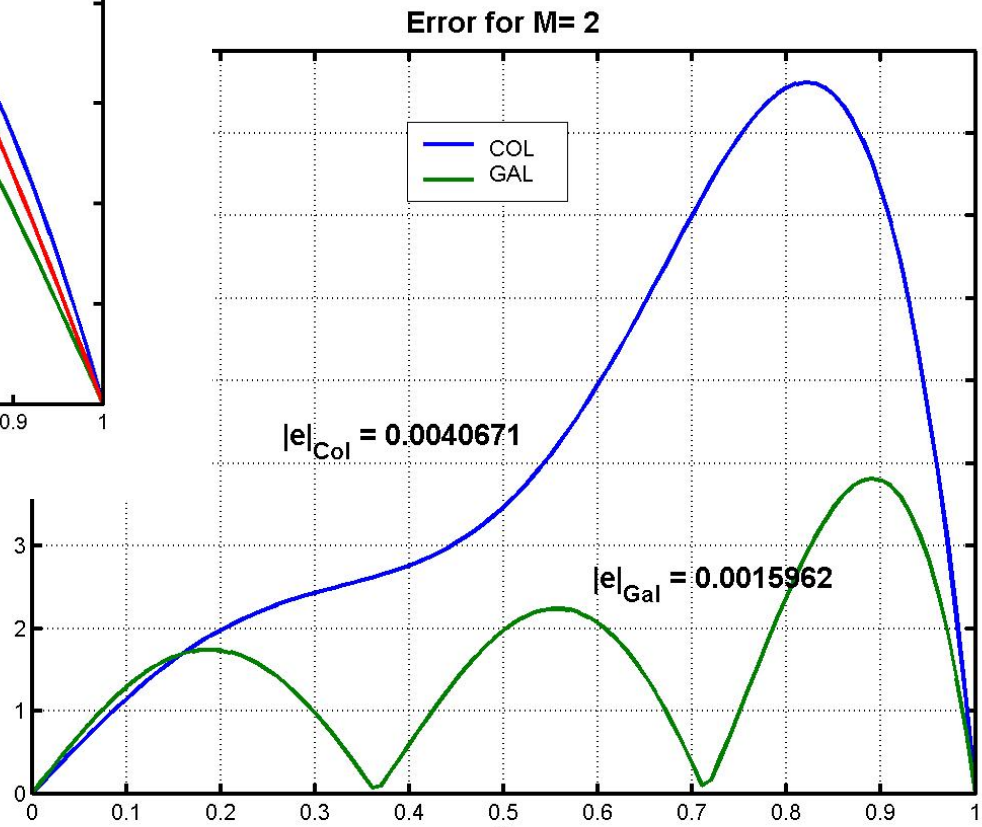
$$\psi = x$$



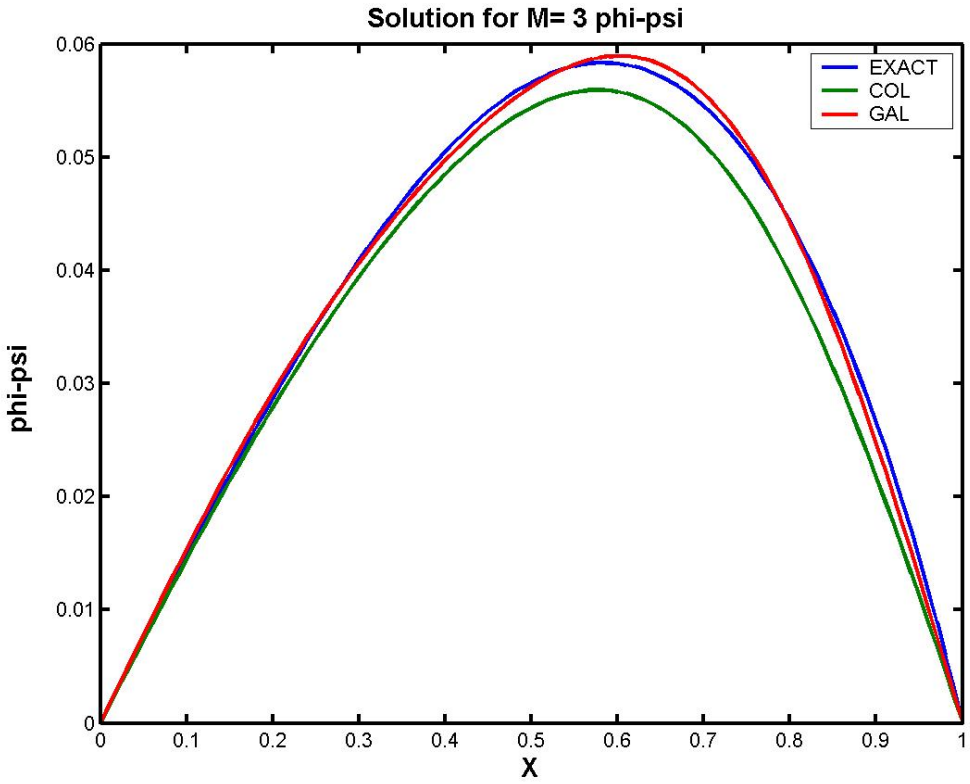
Example 1- Results using point collocation & Galerkin



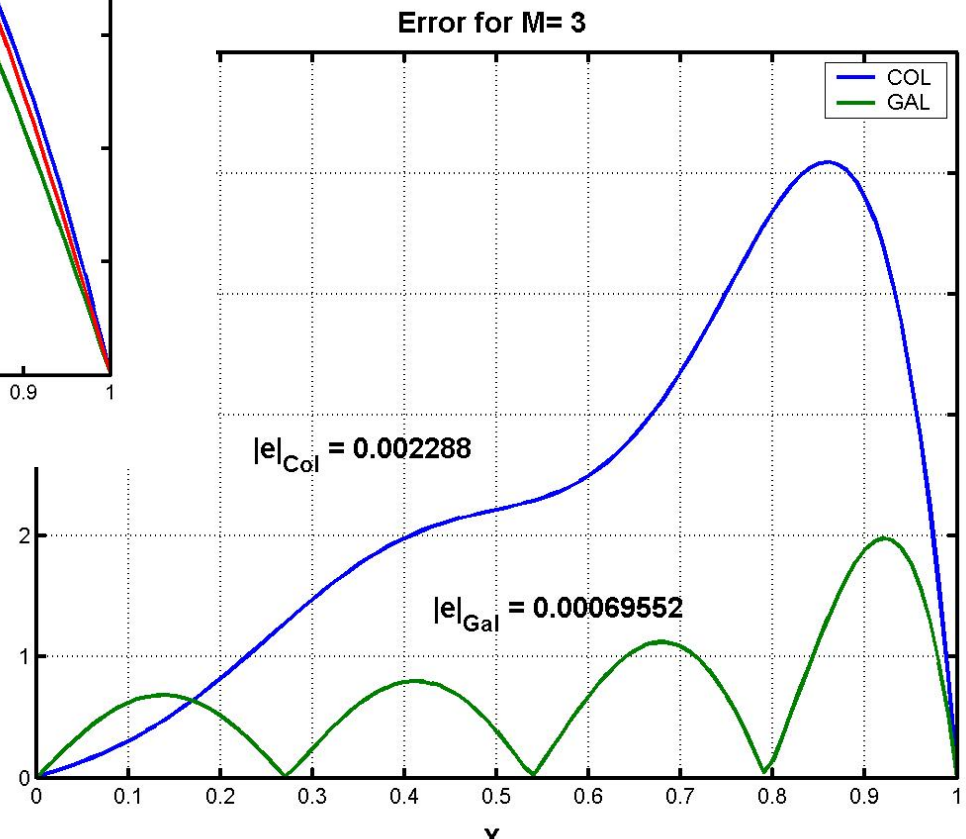
using M = 2



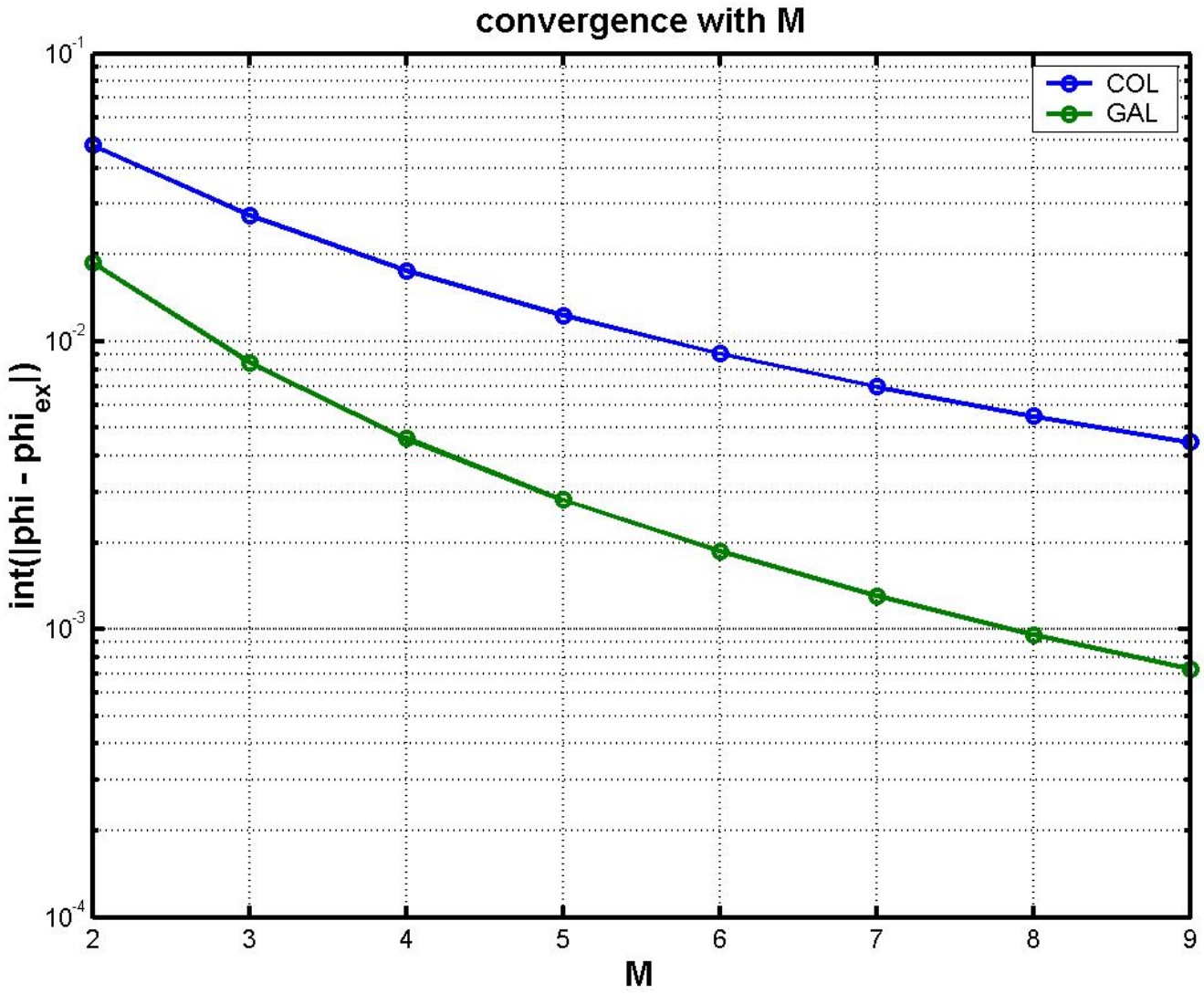
Example 1- Results using point collocation & Galerkin



using M = 3



Example 1- Results using point collocation & Galerkin



Example 2- A 2D scalar elastic problem - Definition

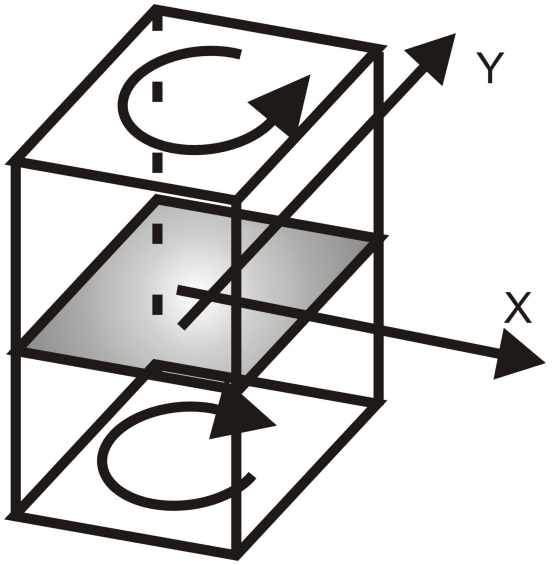
Find $\hat{\phi}(x)$ solution of the following PDE

$$\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}=-2G\theta \qquad \text{in } \Omega:\{(x,y);-3\le x\le 3;-2\le y\le 2\}$$

$$\phi(x=-3)=\phi(x=3)=0$$

$$\phi(y=-2)=\phi(y=2)=0$$

$$G\theta=1 \qquad G:\text{shear modulus};\ \theta:\text{torsion angle}$$



$$L(\phi)=\frac{\partial^2\phi}{\partial x^2}+\frac{\partial^2\phi}{\partial y^2}$$

$$p=2$$

$$M(\phi)=\phi \quad ; \quad r=0$$

$$\text{on } (x=-3)\cup(x=3)\cup(y=-2)\cup(y=2) \quad \text{DIRICHLET}$$

Example 2- Choosing the approximation

$$\phi(x,y) \cong \hat{\phi}(x,y) = \psi(x,y) + \sum_m a_m N_m(x,y)$$

$$\psi(x,y) = 0 \quad \text{satisfying the boundary condition}$$

$$N_m(x,y) = \cos(\alpha_x(m)\frac{\pi\, x}{6})\cos(\alpha_y(m)\frac{\pi\, y}{4})$$

$$\alpha = (\alpha_x, \alpha_y) = \begin{pmatrix} 1 & 1 \\ 3 & 1 \\ 1 & 3 \end{pmatrix}$$

$$N_m(x,y) = 0 \qquad \forall (x,y) \in \Gamma$$

$$R_\Omega = L\hat{\phi} + p = L\psi + \sum_m a_m L N_m + p$$

$$= \sum_m a_m L N_m + p$$

$$L N_m = \frac{\partial^2 N_m}{\partial x^2} + \frac{\partial^2 N_m}{\partial y^2} = -\pi^2 \left(\left(\frac{\alpha_x}{6} \right)^2 + \left(\frac{\alpha_y}{4} \right)^2 \right) N_m$$

Example 2- Weighted residual using Galerkin (see routine Ej_2_2.m)

$$\int_\Omega W_l R_\Omega d\Omega = \int_\Omega W_l \left(\mathsf{L}\hat{\phi} + p \right) d\Omega = 0$$

$$\int_\Omega N_l \left(\sum_m a_m \mathsf{L} N_m + p \right) d\Omega = 0$$

$$\int_\Omega N_l \left(\sum_m a_m \left(- \pi^2 \left(\left(\frac{\alpha_x(m)}{6} \right)^2 + \left(\frac{\alpha_y(m)}{4} \right)^2 \right) N_m \right) + p \right) d\Omega = 0$$

$$K_{lm}\, a_m = f_l$$

$$K_{lm} = - \pi^2 \left(\left(\frac{\alpha_x(m)}{6} \right)^2 + \left(\frac{\alpha_y(m)}{4} \right)^2 \right) \int_\Omega N_l N_m d\Omega$$

$$= - \pi^2 \left(\left(\frac{\alpha_x(m)}{6} \right)^2 + \left(\frac{\alpha_y(m)}{4} \right)^2 \right) \int_2^2 dy \int_3^3 N_l N_m dx$$

$$f_l = - \int_\Omega N_l\, p\, d\Omega = - 2 \int_\Omega N_l\, d\Omega = - 2 \int_2^2 dy \int_3^3 N_l\, dx$$

Example 2- Weighted residual using Galerkin (see routine Ej_2_2_symb.m with symbolic soft)

```
syms ax ay x y

ax=[1,3,1]
ay=[1,1,3]

N = [ cos(ax(1)*pi*x/6)*cos(ay(1)*pi*y/4);
      cos(ax(2)*pi*x/6)*cos(ay(2)*pi*y/4);
      cos(ax(3)*pi*x/6)*cos(ay(3)*pi*y/4)];

for l=1:3,
    for m=1:3,
        factor =    pi^2*((ax(m)/6)^2+(ay(m)/4)^2);
        K(l,m) = factor*int(int(N(l)*N(m),x,-3,3),y,-2,2);
    end
    f(l) = -2*int(int(N(l),x,-3,3),y,-2,2);
end
```

Example 2- Weighted residual using Galerkin (see routine Ej_2_2_num.m)

```
global X Y ll mm coef

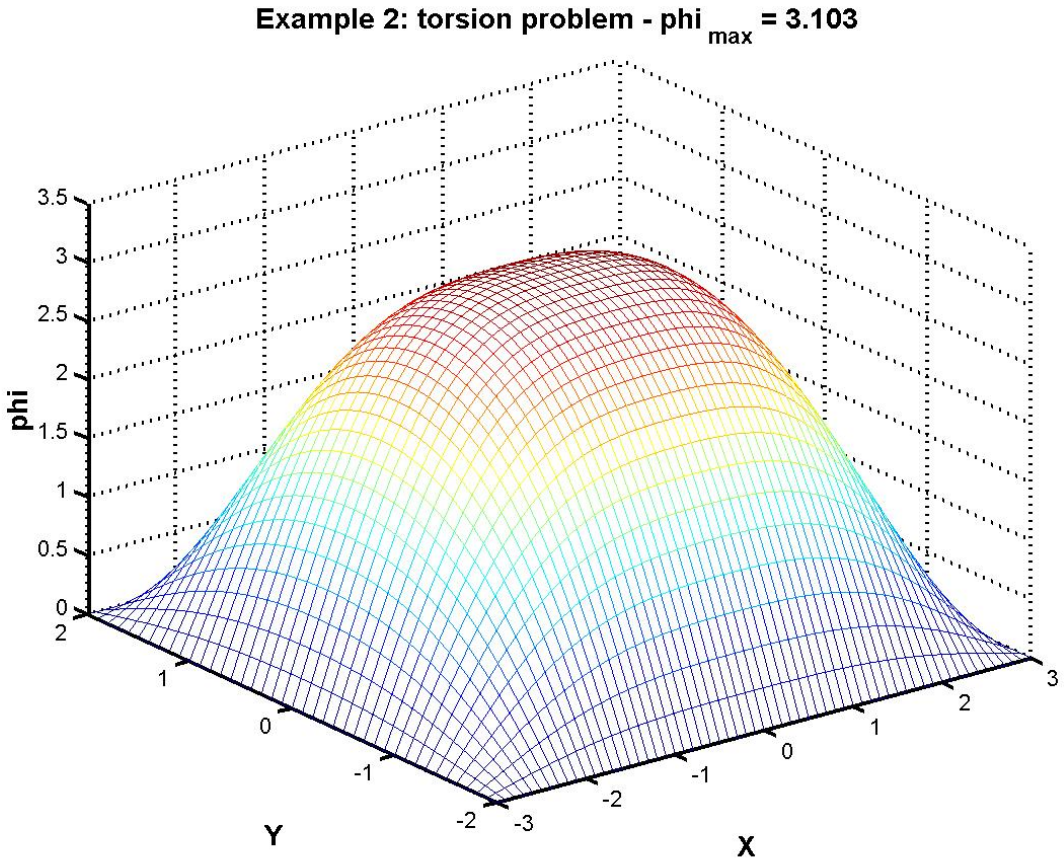
% Number of terms to be used (M)
M = 3;
coef.p = 2;
% grid of samples
[X,Y]=meshgrid(-3:0.1:3,-2:0.1:2);

f = zeros(M,1);
K = zeros(M,M);
for ll=1:M
    f(ll) = gauss_integration('ffun_Ej_2_2_num_rhs',-3,3,-2,2);
    for mm=1:M
        K(ll,mm) = gauss_integration('ffun_Ej_2_2_num_lhs',-3,3,-2,2);

    end
end

a = K\f;
```

Example 2- Weighted residual using Galerkin Results



$$a_1 = \frac{4608}{13\pi^4} \quad ; \quad a_2 = -\frac{4608}{135\pi^4} \quad ; \quad a_3 = -\frac{4608}{255\pi^4}$$

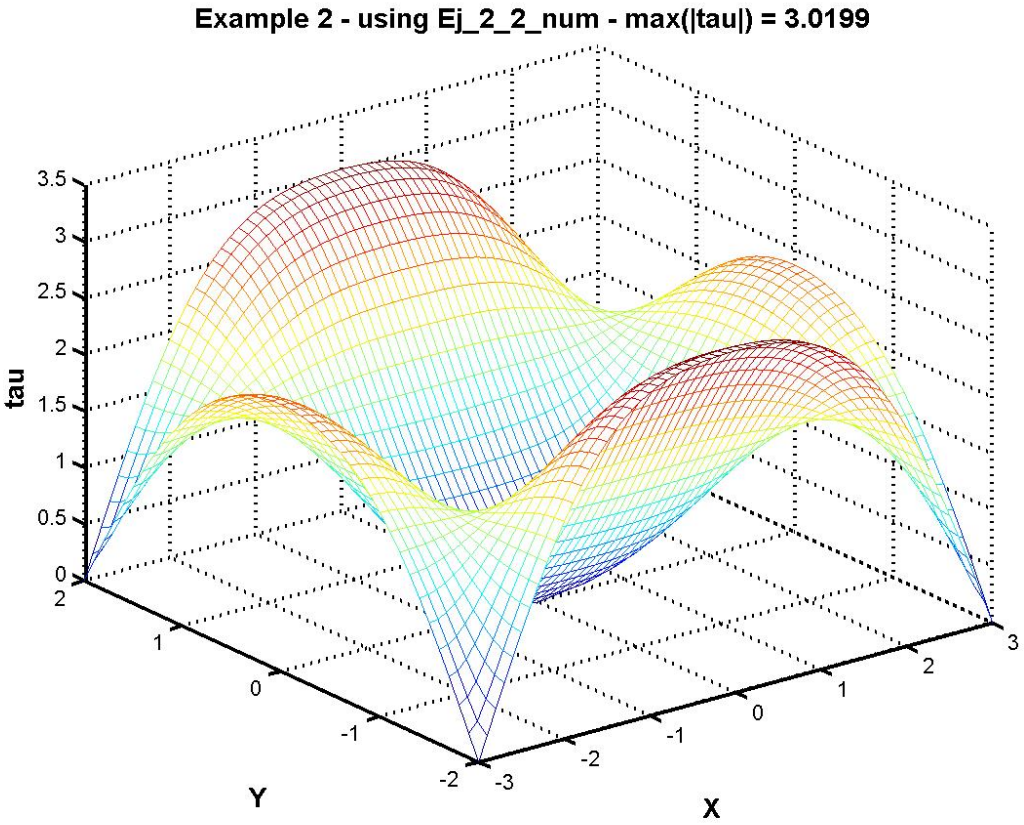
$$T = 2\int_2^2 dy \int_3^3 \hat{\phi}\, dx = 74.265 \qquad \text{(exact value } 76.4\text{)}$$

Example 2- Weighted residual using Galerkin Results

$$\tau_k = \frac{\partial \phi}{\partial x_k} \quad ; \quad |\tau| = \sqrt{\left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2}$$

$$|\hat{\tau}| = \sqrt{\left(\frac{\partial \hat{\phi}}{\partial x}\right)^2 + \left(\frac{\partial \hat{\phi}}{\partial y}\right)^2} = \sqrt{\left(\sum_m a_m \frac{\partial N_m}{\partial x}\right)^2 + \left(\sum_m a_m \frac{\partial N_m}{\partial y}\right)^2}$$

3.02 (exact value 2.96)



Least square approximation to PDE

$$I(a_1, a_2, a_3, \cdots, a_M) = \int_\Omega R_\Omega{}^2 d\Omega$$

$$\frac{\partial I}{\partial a_l} = 0 \qquad l = 1, 2, \cdots, M$$

$$\frac{\partial \int_\Omega R_\Omega{}^2 d\Omega}{\partial a_l} = \int_\Omega \frac{\partial}{\partial a_l} R_\Omega{}^2 d\Omega = \int_\Omega 2 R_\Omega \frac{\partial R_\Omega}{\partial a_l} d\Omega = 0$$

$$\int_\Omega R_\Omega \frac{\partial R_\Omega}{\partial a_l} d\Omega = 0 \quad \Rightarrow W_l = \frac{\partial R_\Omega}{\partial a_l} = \mathrm{L} N_l$$

$$\therefore \int_\Omega \mathrm{L} N_l \; R_\Omega = 0$$

LEAST SQUARE APPROXIMATION $\neq$ GALERKIN

Galerkin Least Square approximation to PDE (GLS)

$$\int_{\Omega} \mathrm{L}N_l \; R_{\Omega} = 0$$

$$\int_{\Omega} \mathrm{L}N_l \left\{ \left(\mathrm{L}\psi + \sum_m a_m \mathrm{L}N_m \right) + p \right\} = 0$$

$$\int_{\Omega} \mathrm{L}N_l \sum_m a_m \, \mathrm{L}N_m \, d\Omega = - \int_{\Omega} \mathrm{L}N_l \left(\mathrm{L}\psi + p \right) d\Omega$$

$$K_{lm} = \int_{\Omega} \mathrm{L}N_l \, \mathrm{L}N_m \, d\Omega$$

$$f_l = - \int_{\Omega} \mathrm{L}N_l \left(\mathrm{L}\psi + p \right) d\Omega$$

GALERKIN LEAST SQUARE APPROXIMATION

Simultaneous approximation to the solution of the PDE in the domain and on its boundary.

$$R_\Omega = A(\hat{\phi}) = \mathsf{L}(\hat{\phi}) + \mathrm{p} = 0 \qquad \text{in } \Omega$$

$$R_\Gamma = B(\hat{\phi}) = \mathsf{M}(\hat{\phi}) + \mathrm{r} = 0 \qquad \text{in } \Gamma$$

$$\phi(x) \cong \hat{\phi}(x) = \sum_m a_m N_m(x) \qquad \text{note that } \psi \text{ does not exist}$$

$$\int_\Omega W_l \, R_\Omega \, d\Omega + \int_\Gamma \overline{W}_l \, R_\Gamma \, d\Gamma = 0$$

Simultaneous approximation to the solution of the PDE in the domain and on its boundary.

$$\int_{\Omega} W_l \; R_{\Omega} \; d\Omega + \int_{\Gamma} \overline{W}_l \; R_{\Gamma} \; d\Gamma = 0$$

$$\int_{\Omega} W_l \left(\sum_m a_m \mathrm{L} N_m + p \right) d\Omega + \int_{\Gamma} \overline{W}_l \left(\sum_m a_m \mathrm{M} N_m + r \right) d\Gamma = 0$$

$$\int_{\Omega} W_l \left(\sum_m a_m \mathrm{L} N_m \right) d\Omega + \int_{\Gamma} \overline{W}_l \left(\sum_m a_m \mathrm{M} N_m \right) d\Gamma = - \int_{\Omega} W_l \, p \; d\Omega - \int_{\Gamma} \overline{W}_l \; r \; d\Gamma$$

$$K_{lm} a_m = f_l$$

$$K_{lm} = \int_{\Omega} W_l \mathrm{L} N_m \; d\Omega + \int_{\Gamma} \overline{W}_l \mathrm{M} N_m \; d\Gamma$$

$$f_l = - \int_{\Omega} W_l \; p \; d\Omega - \int_{\Gamma} \overline{W}_l \; r \; d\Gamma \qquad l, m = 1, 2, \cdots, M$$

Example 3 = Example 1- but solved without psi ψ

Find $\hat{\phi}(x)$ solution of the following ODE

$$\frac{d^2\phi}{dx^2} - \phi = 0 \qquad \text{in } \Omega : \{ x; 0 \le x \le 1 \}$$

$$\phi(x = 0) = 0$$

$$\phi(x = 1) = 1 \qquad \text{in } \Gamma : \{ x = 0, x = 1 \}$$

$$L(\phi) = \frac{d^2\phi}{dx^2} - \phi$$

$$p = 0$$

$$M(\phi) + r = 0 \qquad \text{in } \Gamma$$

$$\begin{cases} M(\phi) = \phi & ; \quad r = 0 \quad \text{on } x = 0 \quad \text{DIRICHLET} \\ M(\phi) = \phi & ; \quad r = -1 \quad \text{on } x = 1 \quad \text{DIRICHLET} \end{cases}$$

Example 3 – weighted residual formulation

$$\int_{\Omega} W_l \, R_{\Omega} \, d\Omega + \int_{\Gamma} \overline{W}_l \, R_{\Gamma} \, d\Gamma = 0$$

$$\int_0^1 W_l R_{\Omega} \, dx + \left[\overline{W}_l R_{\Gamma} \right]_{x=0} + \left[\overline{W}_l R_{\Gamma} \right]_{x=1} = 0$$

$$W_l = N_l \quad \text{and} \quad \overline{W}_l = - N_l \big|_{\Gamma}$$

$$\int_0^1 N_l \left(\frac{d^2 \hat{\phi}}{dx^2} - \hat{\phi} \right) dx - \left[N_l \hat{\phi} \right]_{x=0} - \left[N_l \left(\hat{\phi} - 1 \right) \right]_{x=1} = 0$$

using $N_m = x^{m-1}, \; m = 1,2,3,\cdots$

$$K_{lm} a_m = f_l$$

$$K_{lm} = \int_0^1 N_l \left(\frac{d^2 N_m}{dx^2} - N_m \right) dx - \left[N_l N_m \right]_{x=0} - \left[N_l N_m \right]_{x=1}$$

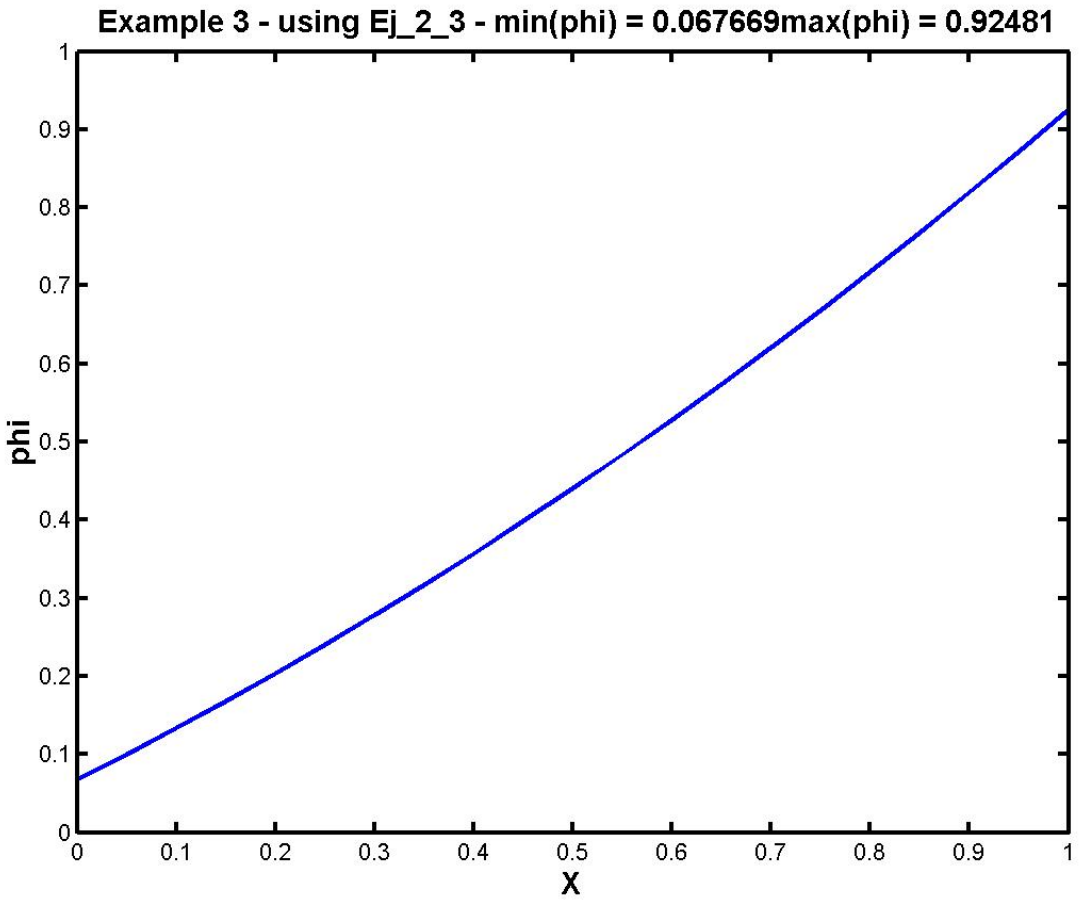
$$f_l = - \left[N_l \right]_{x=1} \qquad l, m = 1,2,\cdots, M$$

Example 3 – algebraic system

$$K\,a = f$$

$$K = -\begin{vmatrix} 3 & \frac{3}{2} & -\frac{2}{3} \\ \frac{3}{2} & \frac{4}{3} & \frac{1}{4} \\ \frac{4}{3} & \frac{5}{4} & \frac{8}{15} \end{vmatrix}$$

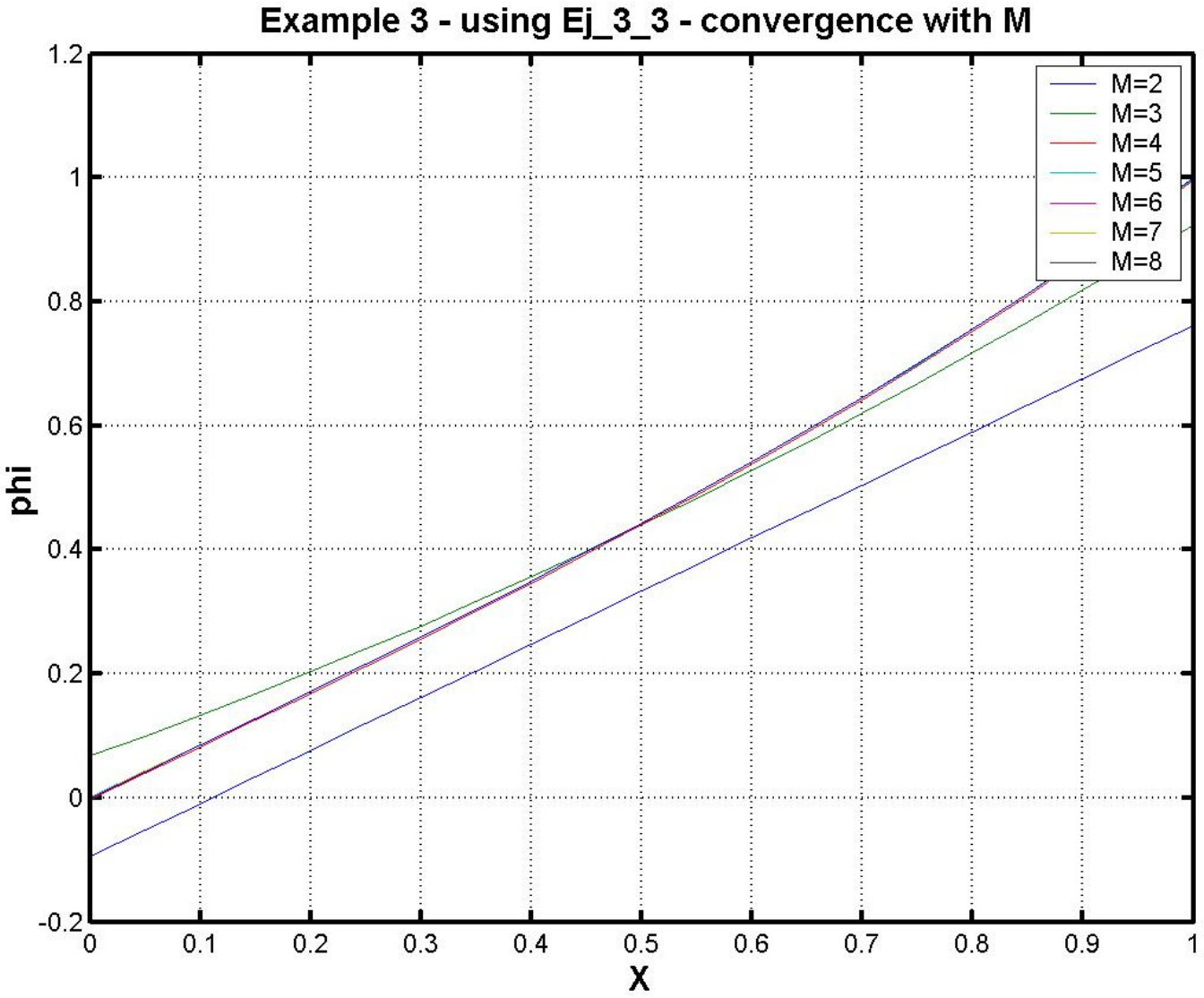
$$f = -\begin{vmatrix} 1 & 1 & 1 \end{vmatrix}^{T}$$



Example 3 – Results

M	Phi (x = 0)	Phi (x = 1)
2.0000e+000 -	9.5238e-002	7.6190e-001
3.0000e+000	6.7669e-002	9.2481e-001
4.0000e+000	-3.1787e-003	9.9566e-001
5.0000e+000	5.6585e-004	9.9940e-001
6.0000e+000	-1.3409e-005	9.9998e-001
7.0000e+000	1.4329e-006	1.0000e+000

Example 3 – algebraic system



Example 4 = Example 2- but solved without ψ

$$N_j = \underbrace{(4 - y^2)}_{\text{satisfy } M(\phi)+r=0\big|_{y=\pm 2}} \times N_j^*$$

$$N_1^* = 1 \quad ; \quad N_2^* = x^2 \quad ; \quad N_3^* = y^2$$

$$N_4^* = x^2 y^2 \quad ; \quad N_5^* = x^4 \quad \text{and so on}$$

$$\hat{\phi} = (4 - y^2)(a_1 + a_2 x^2 + a_3 y^2 + a_4 x^2 y^2 + a_5 x^4)$$

$$\int_{-3}^{3}\int_{-2}^{2} W_l \left(\frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} + 2 \right) dy\, dx + \int_{-2}^{2} \overline{W_l}\, \hat{\phi}\Big|_{x=3} dy - \int_{2}^{-2} \overline{W_l}\, \hat{\phi}\Big|_{x=-3} dy = 0$$

Natural boundary conditions

$$\int_\Omega W_l \, R_\Omega \, d\Omega = \int_\Omega W_l \left(\mathsf{L}\hat{\phi} + \mathrm{p} \right) d\Omega = 0$$

integration by parts allows to reduce the smoothness requirement over the functional space

integration by parts

$$\int_\Omega W_l \left(\mathsf{L}\hat{\phi} + \mathrm{p} \right) d\Omega = \int_\Omega \mathsf{C} W_l \, \mathsf{D}\,\hat{\phi} \, d\Omega + \int_\Gamma W_l \, \mathsf{E}\,\hat{\phi} \, d\Gamma + \int_\Omega W_l \; p d\Omega = 0$$

strong form — weak form

with $\mathsf{C}, \mathsf{D}, \mathsf{E}$ linear operators of lower order than L

Natural boundary conditions

integration by parts

$$\int_\Omega W_l \left(\mathrm{L}\hat{\phi} + \mathrm{p}\right) d\Omega = \int_\Omega \mathrm{C} W_l \,\mathrm{D}\,\hat{\phi}\, d\Omega + \int_\Gamma W_l \,\mathrm{E}\,\hat{\phi}\, d\Gamma + \int_\Omega W_l \; p d\Omega = 0$$

strong form

weak form

$$\int_\Omega W_l \left(\mathrm{L}\hat{\phi} + \mathrm{p}\right) d\Omega + \int_\Gamma \overline{W_l} \left(\mathrm{M}\;\hat{\phi} + \mathrm{r}\right) d\Gamma =$$

$$= \int_\Omega \mathrm{C} W_l \,\mathrm{D}\,\hat{\phi}\, d\Omega + \int_\Omega W_l \; p d\Omega + \int_\Gamma W_l \,\mathrm{E}\,\hat{\phi}\, d\Gamma + \int_\Gamma \overline{W_l} \left(\mathrm{M}\;\hat{\phi} + \mathrm{r}\right) d\Gamma = 0$$

combining the last two boundary integrals, is it possible to cancel out ?

This is only possible for certain selections of $\overline{W_l}$ and boundary conditions

Example 5 – natural boundary conditions

Find $\hat{\phi}(x)$ solution of the following ODE

$$\frac{\mathrm{d}^2\phi}{\mathrm{dx}^2} - \phi = 0 \qquad \text{in } \Omega : \{ \mathrm{x}; 0 \le \mathrm{x} \le 1 \}$$

$$\phi(x = 0) = 0$$

$$\frac{d\phi}{dx}(x = 1) = 20 \qquad \text{in } \Gamma : \{ \mathrm{x} = 0, \mathrm{x} = 1 \}$$

using $N_m = x^m$

$$\int_0^1 W_l R_\Omega \, dx + \left[\overline{W}_l R_\Gamma \right]_{x=0} + \left[\overline{W}_l R_\Gamma \right]_{x=1} = 0$$

$$\int_0^1 W_l \left(\frac{d^2\hat{\phi}}{dx^2} - \hat{\phi} \right) dx + \left[\overline{W_l} \hat{\phi} \right]_{x=0} + \left[\overline{W_l} \left(\frac{d\hat{\phi}}{dx} - 20 \right) \right]_{x=1} = 0$$

$$- \int_0^1 \frac{dW_l}{dx} \frac{d\hat{\phi}}{dx} dx - \int_0^1 W_l \, \hat{\phi} \, dx + \left[W \frac{d\hat{\phi}}{dx} \right]_{x=1} - \left[W \frac{d\hat{\phi}}{dx} \right]_{x=0} +$$

$$\left[\overline{W_l} \hat{\phi} \right]_{x=0} + \left[\overline{W_l} \left(\frac{d\phi}{dx} - 20 \right) \right]_{x=1} = 0$$

$\left[\overline{W_l} \hat{\phi} \right]_{x=0} = 0$ (DIRICHLET)

Example 5 – natural boundary conditions

Choosing $\overline{W}_l\Big|_{x=1} = -W_l\Big|_{x=1}$ & $W_l\Big|_{x=0} = 0$

$$-\int_0^1 \frac{dW_l}{dx}\frac{d\hat{\phi}}{dx}dx - \int_0^1 W_l\,\hat{\phi}\,dx + \left[W_l\,\frac{d\hat{\phi}}{dx}\right]_{x=1} - \left[W_l\,\frac{d\hat{\phi}}{dx}\right]_{x=0} +$$

$$\underbrace{\left[\overline{W}_l\hat{\phi}\right]_{\Gamma_D}}_{=0\,(\text{DIRICHLET})} + \left[\overline{W}_l\left(\frac{d\hat{\phi}}{dx} - 20\right)\right]_{x=1} = 0$$

$$\Rightarrow \qquad \int_0^1 \frac{dW_l}{dx}\frac{d\hat{\phi}}{dx}dx + \int_0^1 W_l\,\hat{\phi}\,dx = \left[20\,W_l\right]_{x=1}$$

there is no need to compute the $\left.\frac{d\hat{\phi}}{dx}\right|_{\Gamma}$

Gradient boundary conditions are natural

for such weak formulation

Moreover, if $\left.\frac{d\hat{\phi}}{dx}\right|_{\Gamma} = 0 \Rightarrow rhs = 0$

Example 5 – natural boundary conditions

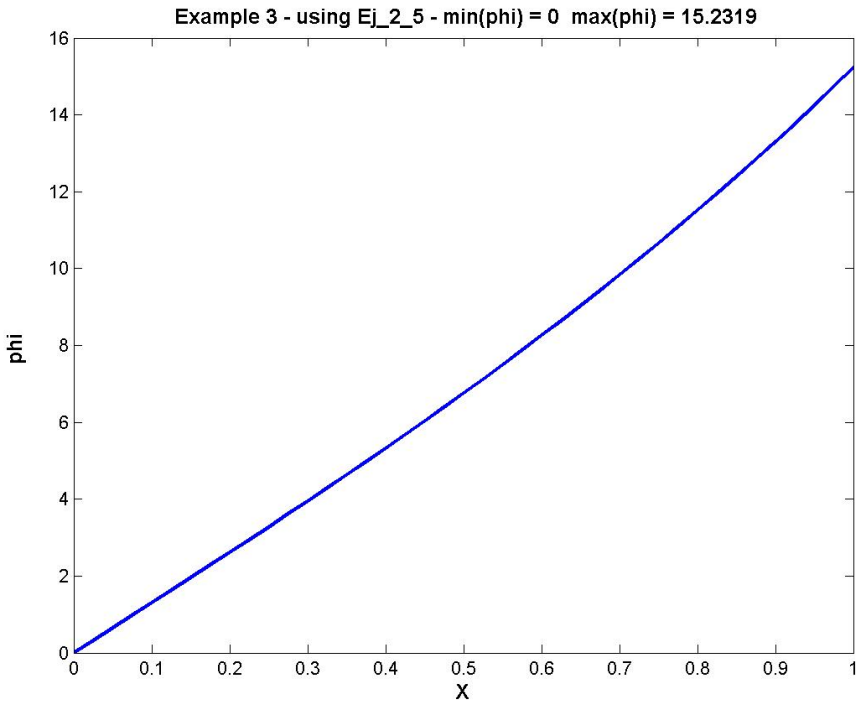
$$\int_0^1 \frac{dW_l}{dx}\frac{d\hat{\phi}}{dx}\,dx + \int_0^1 W_l\,\hat{\phi}\,dx = \left[20\,W_l\right]_{x=1}$$

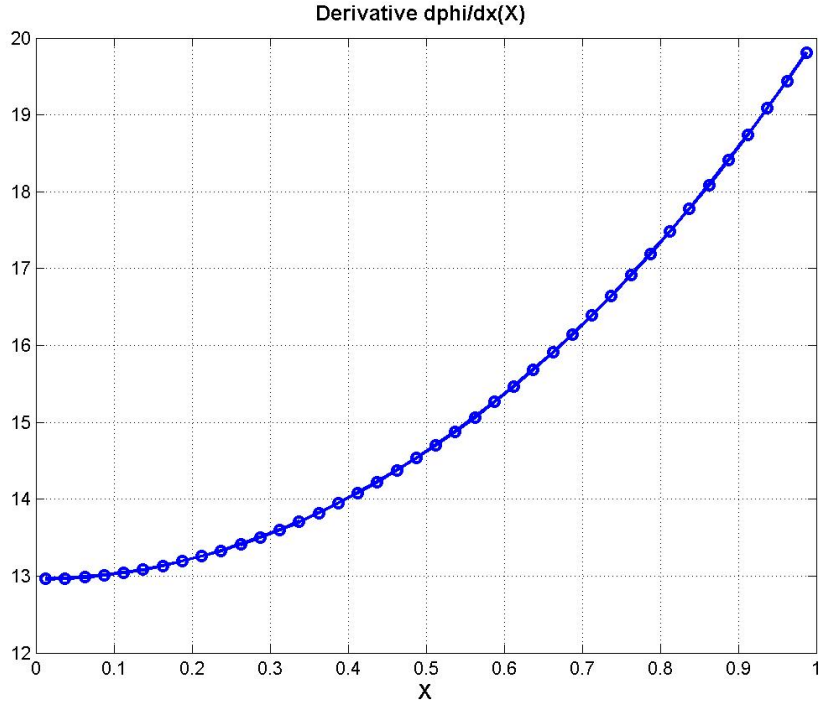
$$\Rightarrow \quad K\,a = f$$

$$K_{lm} = \int_0^1 \frac{dW_l}{dx}\frac{dN_m}{dx}\,dx + \int_0^1 W_l\,N_m\,dx$$

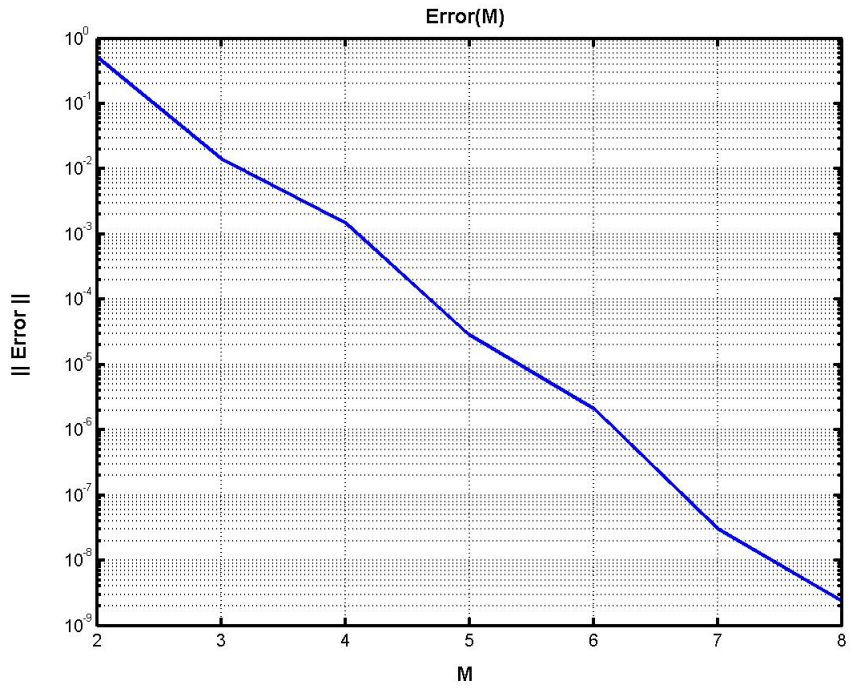
$$f_l = 20\,W_l\Big|_{x=1}$$

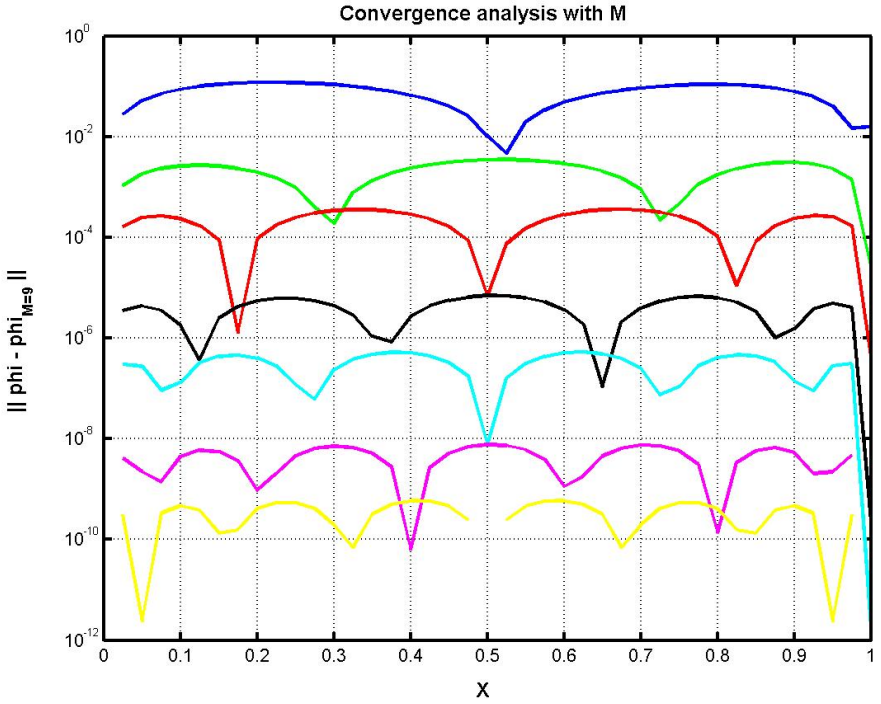
Example 5 – Results (routine Ej_2_5.m)





Example 5 – Results (routine Ej_2_5.m)





Natural boundary condition for heat equation

Find $\hat{\phi}(x)$ solution of the following PDE

$$\frac{\partial}{\partial \mathrm{x}}\left(\kappa \frac{\partial \phi}{\partial \mathrm{x}}\right)+\frac{\partial}{\partial \mathrm{y}}\left(\kappa \frac{\partial \phi}{\partial \mathrm{y}}\right)+Q=0 \qquad \text{in } \Omega:\left\{\mathrm{x}; x \in \Re^{2}\right\}$$

$$\phi=\bar{\phi} \quad \text{on } \Gamma_{\phi}$$

$$\kappa \frac{\partial \phi}{\partial \eta}=-\bar{q} \quad \text{on } \Gamma_{q}$$

$$\hat{\phi}=\psi+\sum_{m} a_{m} N_{m}$$

$$\psi=\bar{\phi} \quad \text{on } \Gamma_{\phi}$$

$$N_{m}=0 \quad \text{on } \Gamma_{\phi}$$

$$\int_{\Omega} W_{l}\left(\frac{\partial}{\partial \mathrm{x}}\left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{x}}\right)+\frac{\partial}{\partial \mathrm{y}}\left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{y}}\right)+Q\right) d\Omega+\int_{\Gamma_{q}} \overline{W_{l}}\left(\kappa \frac{\partial \hat{\phi}}{\partial \eta}+\bar{q}\right) d\Gamma=0$$

Natural boundary condition for heat equation

$$\int_{\Omega} W_l \left(\frac{\partial}{\partial \mathrm{x}} \left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{x}} \right) + \frac{\partial}{\partial \mathrm{y}} \left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{y}} \right) + Q \right) d\Omega + \int_{\Gamma_q} \overline{W_l} \left(\kappa \frac{\partial \hat{\phi}}{\partial \eta} + \bar{q} \right) d\Gamma = 0$$

Applying the Green's lemma

$$\int_{\Omega} \left\{ - \left(\frac{\partial W_l}{\partial \mathrm{x}} \left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{x}} \right) + \frac{\partial W_l}{\partial \mathrm{y}} \left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{y}} \right) \right) + W_l Q \right\} d\Omega +$$

$$+ \int_{\Gamma_\phi \cup \Gamma_q} W_l \left(\kappa \frac{\partial \hat{\phi}}{\partial \eta} + \bar{q} \right) d\Gamma + \int_{\Gamma_q} \overline{W_l} \left(\kappa \frac{\partial \hat{\phi}}{\partial \eta} + \bar{q} \right) d\Gamma = 0$$

choosing the weighted functions so that

$$W_l = 0 \quad \text{on} \quad \Gamma_\phi$$

$$\overline{W_l} = - W_l \quad \text{on} \quad \Gamma_q$$

$$\int_{\Omega} \left(\frac{\partial W_l}{\partial \mathrm{x}} \left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{x}} \right) + \frac{\partial W_l}{\partial \mathrm{y}} \left(\kappa \frac{\partial \hat{\phi}}{\partial \mathrm{y}} \right) - W_l Q \right) d\Omega + \int_{\Gamma_q} W_l \bar{q} \, d\Gamma = 0$$

Natural boundary condition for heat equation

$$K\,a = f$$

$$K_{lm} = \int_{\Omega} \left(\frac{\partial W_l}{\partial \mathrm{x}} \left(\kappa \frac{\partial N_m}{\partial \mathrm{x}} \right) + \frac{\partial W_l}{\partial \mathrm{y}} \left(\kappa \frac{\partial N_m}{\partial \mathrm{y}} \right) \right) d\Omega$$

$$f_l = \int_{\Omega} W_l Q \, d\Omega + \int_{\Gamma_q} W_l \bar{q} \, d\Gamma - \int_{\Omega} \left(\frac{\partial W_l}{\partial \mathrm{x}} \left(\kappa \frac{\partial \psi}{\partial \mathrm{x}} \right) + \frac{\partial W_l}{\partial \mathrm{y}} \left(\kappa \frac{\partial \psi}{\partial \mathrm{y}} \right) \right) d\Omega = 0$$

$$1 \le l, m \le M$$

using Galerkin approximation ($W_l = N_l$)

$\therefore \quad K_{lm} = K_{ml}$ symmetry

$\kappa \frac{\partial \phi}{\partial \eta} = -\bar{q}$ on $\Gamma_q \quad \Rightarrow$ there is no phi gradient at boundary terms

$\therefore \quad \kappa \frac{\partial \phi}{\partial \eta} = -\bar{q}$ is the natural BC for heat equation

Example 6 – natural BC for heat equation

Find $\hat{\phi}(x)$ solution of the following PDE in 2D

$$\frac{\partial}{\partial x}\left(\frac{\partial \phi}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{\partial \phi}{\partial y}\right)=0 \qquad \text{in } \Omega:\{x, y; x, y \in (-1,1)\times(-1,1)\}$$

$$\phi = 0 \quad \text{on } \Gamma_\phi:\{x, y; y = \pm 1\}$$

$$\frac{\partial \phi}{\partial \eta}=\cos\left(\frac{\pi y}{2}\right) \quad \text{on } \Gamma_q:\{x, y; x = \pm 1\}$$

$$N_j = \underset{\text{satisfy } M(\phi)+r=0|_{y=\pm 1}}{\left(1-y^2\right)} \times N_j^*$$

$$N_1^* = 1 \quad ; \quad N_2^* = x^2 \quad ; \quad N_3^* = y^2$$

$$N_4^* = x^2 y^2 \quad ; \quad N_5^* = x^4 \quad \text{and so on}$$

$$\hat{\phi} = \left(1-y^2\right)\left(a_1 + a_2 x^2 + a_3 y^2 + a_4 x^2 y^2 + a_5 x^4\right)$$

immediately satisfies the BC @ Γ_ϕ

$$\int_{-1}^{1}\int_{-1}^{1} W_l\left(\frac{\partial^2 \hat{\phi}}{\partial x^2}+\frac{\partial^2 \hat{\phi}}{\partial y^2}\right) dy\, dx + \int_{\Gamma_q} \overline{W_l}\left(\frac{\partial \hat{\phi}}{\partial \eta}-\cos\left(\frac{\pi y}{2}\right)\right) d\Gamma = 0$$

Example 6 – weak form

$$\int_{-1}^{1}\int_{-1}^{1} W_l \left(\frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} \right) dy\, dx + \int_{\Gamma_q} \overline{W_l} \left(\frac{\partial \hat{\phi}}{\partial \eta} - \cos\left(\frac{\pi\, y}{2} \right) \right) d\Gamma = 0$$

$$\int_{-1}^{1}\int_{-1}^{1} \left(\frac{\partial W_l}{\partial x} \frac{\partial \hat{\phi}}{\partial x} + \frac{\partial W_l}{\partial y} \frac{\partial \hat{\phi}}{\partial y} \right) dy\, dx -$$

$$- \int_{\Gamma_q \cup \Gamma_\phi} W_l \frac{\partial \hat{\phi}}{\partial \eta}\, d\Gamma - \int_{\Gamma_q} \overline{W_l} \left(\frac{\partial \hat{\phi}}{\partial \eta} - \cos\left(\frac{\pi\, y}{2} \right) \right) d\Gamma = 0$$

Example 6 – weak form

$W_l = N_l$ (Galerkin)

$$\Rightarrow \int_{\Gamma_\phi} W_l \frac{\partial \hat{\phi}}{\partial \eta} d\Gamma = 0 \quad \text{because } N_l\big|_{\Gamma_\phi} = 0$$

$$\int_{-1}^{1}\int_{-1}^{1} \left(\frac{\partial N_l}{\partial x}\frac{\partial \hat{\phi}}{\partial x} + \frac{\partial N_l}{\partial y}\frac{\partial \hat{\phi}}{\partial y} \right) dy\, dx -$$

$$- \int_{\Gamma_q} N_l \frac{\partial \hat{\phi}}{\partial \eta} d\Gamma - \int_{\Gamma_q} \overline{W_l} \left(\frac{\partial \hat{\phi}}{\partial \eta} - \cos\left(\frac{\pi\, y}{2} \right) \right) d\Gamma = 0$$

taking $\overline{W_l}\big|_{\Gamma_q} = -\overline{N_l}\big|_{\Gamma_q}$ cancel out the derivative term @ Γ_q

$$\therefore \quad \int_{-1}^{1}\int_{-1}^{1} \left(\frac{\partial N_l}{\partial x}\frac{\partial \hat{\phi}}{\partial x} + \frac{\partial N_l}{\partial y}\frac{\partial \hat{\phi}}{\partial y} \right) dy\, dx - \int_{\Gamma_q} N_l \cos\left(\frac{\pi\, y}{2} \right) \left(\frac{\partial \hat{\phi}}{\partial \eta} - \right) d\Gamma = 0$$

Example 6 – linear system

$$\therefore \quad K_{lm} = \int_{-1}^{1}\int_{-1}^{1}\left(\frac{\partial N_l}{\partial x}\frac{\partial \hat{\phi}}{\partial x} + \frac{\partial N_l}{\partial y}\frac{\partial \hat{\phi}}{\partial y} \right) dy\, dx$$

$$f_l = \int_{-1}^{1} N_l\big|_{x=1} \cos\left(\frac{\pi y}{2}\right) dy - \int_{1}^{-1} N_l\big|_{x=-1} \cos\left(\frac{\pi y}{2}\right) dy$$

equivalent to

$$K_{lm} = \int_{0}^{1}\int_{0}^{1}\left(\frac{\partial N_l}{\partial x}\frac{\partial \hat{\phi}}{\partial x} + \frac{\partial N_l}{\partial y}\frac{\partial \hat{\phi}}{\partial y} \right) dy\, dx$$

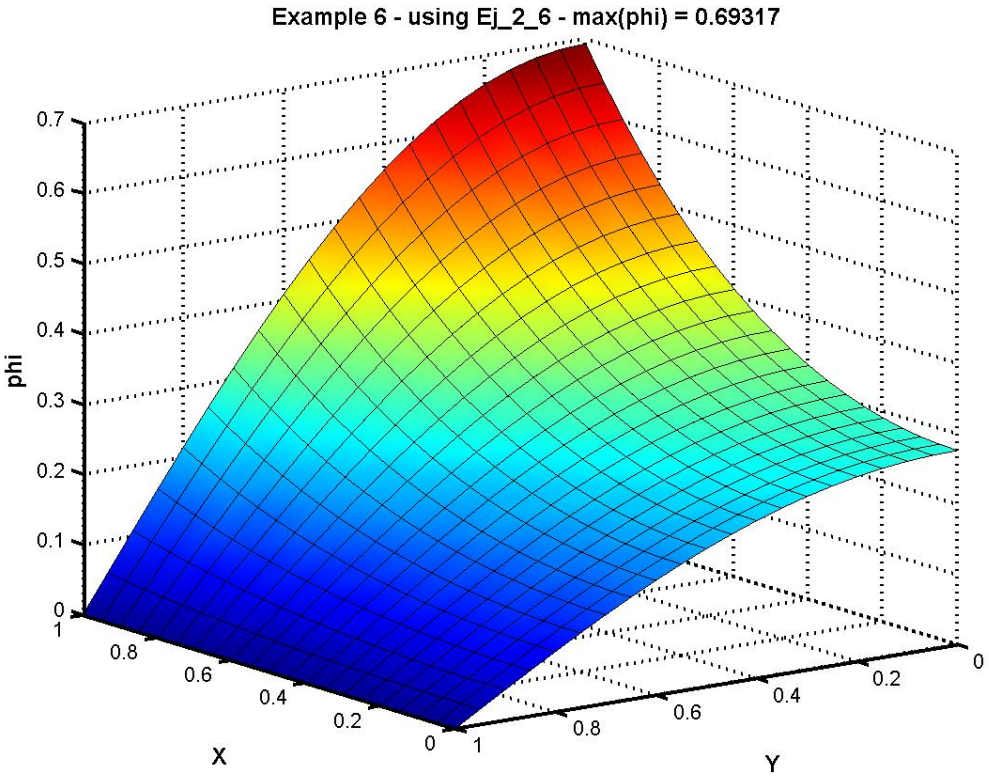
2D integral

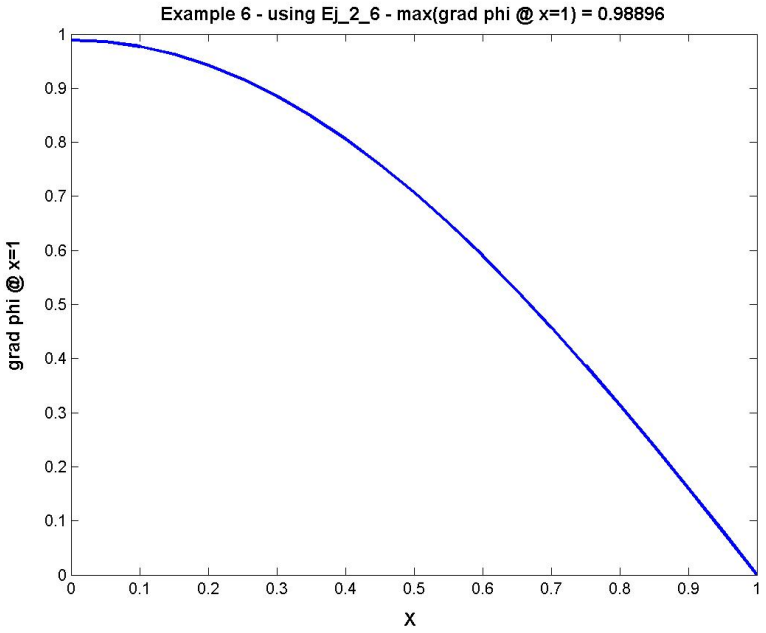
$$f_l = \int_{0}^{1} N_l\big|_{x=1} \cos\left(\frac{\pi y}{2}\right) dy$$

1D integral

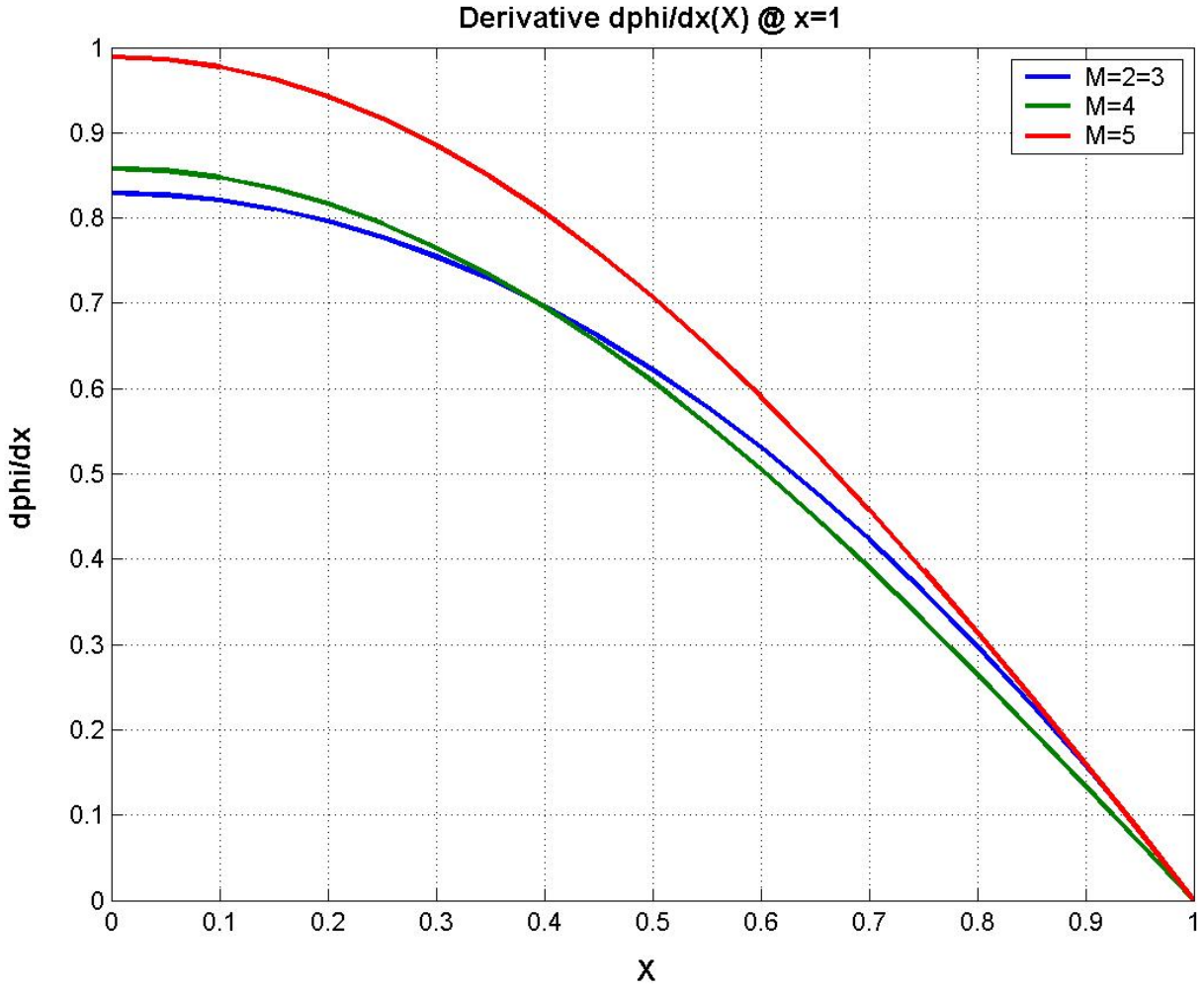
```
f = zeros(M,1);
K = zeros(M,M);
for ll=1:M
    f(ll) =
gauss_integration('ffun_Ej_2_6_rhs',1,c,d);
    for mm=1:M
        K(ll,mm) =
gauss_integration('ffun_Ej_2_6_lhs',2,a,b,c,d);
    end
end
```

Example 6 – Results (routine Ej_2_6.m)





Example 6 – Results (routine Ej_2_6.m)



Example 6 – Results (routine Ej_2_6.m)

K = […
1.3333e+000 4.4444e-001 2.6667e-001 8.8889e-002 2.6667e-001
4.4444e-001 9.7778e-001 8.8889e-002 1.5492e-001 1.0438e+000
2.6667e-001 8.8889e-002 4.1905e-001 1.3968e-001 5.3333e-002
8.8889e-002 1.5492e-001 1.3968e-001 1.1767e-001 1.6000e-001
2.6667e-001 1.0438e+000 5.3333e-002 1.6000e-001 1.3672e+000]

f’ = [5.1602e-001 5.1602e-001 7.0480e-002 7.0480e-002 5.1602e-001]

a’ = [2.7631e-001 3.3925e-001 -5.8747e-002 -9.2206e-002 7.7616e-002]

Boundary solution methods

[1] $\psi(x)$ to satisfy BC

$$\phi(x) \cong \hat{\phi}(x) = \psi(x) + \sum_m a_m N_m(x)$$

$$\text{on } \Gamma \quad \begin{cases} \mathrm{M}(\psi) = -r \\ \mathrm{M}(N_m) = 0 \end{cases} \quad ; \quad m = 1,2,\cdots$$

Weighted Residual excusively over the domain

$$\int_\Omega W_l \, R_\Omega \, d\Omega = \int_\Omega W_l \left(\mathrm{L}\psi + \sum_m a_m \mathrm{L}N_m + p \right) d\Omega = 0$$

[2] give up $\psi(x)$ to satisfy BC and add a boundary residual term in the formulation

$$\phi(x) \cong \hat{\phi}(x) = \sum_m a_m N_m(x)$$

$$R_\Omega = A(\phi) = \mathrm{L}(\phi) + \mathrm{p} = 0 \qquad \text{in } \Omega$$

$$R_\Gamma = B(\phi) = \mathrm{M}(\phi) + \mathrm{r} = 0 \qquad \text{in } \Gamma$$

$$\int_\Omega W_l \, R_\Omega \, d\Omega + \int_\Omega \overline{W}_l \, R_\Gamma \, d\Gamma = 0$$

Boundary solution methods

[1] $\Rightarrow \quad \int_{\Omega} W_l \; R_{\Omega} \; d\Omega = 0$

[2] $\Rightarrow \quad \int_{\Omega} W_l \; R_{\Omega} \; d\Omega + \int_{\Omega} \overline{W}_l \; R_{\Gamma} \; d\Gamma = 0$

why don't propose to solve exclusively the boundary

including for the approximation the solution inside the domain ?

Probably it is difficult to know the analytical solution in the domain

Only for simple (linear) differential operators

But, the dimension of the problem is reduced in one (3D $\rightarrow$ 2D)

[3] $\Rightarrow \quad \int_{\Omega} \overline{W}_l \; R_{\Gamma} \; d\Gamma = 0$

choosing N_m such that $A(N_m) = 0 \quad \therefore \; R_{\Omega} = A(\phi) = \sum_m a_m A(N_m) = 0$

Boundary solution methods to Laplace equation

Any analytic function of the complex variable $z = x + i\,y$

$$f(z) = u + i\,v$$

with $u, v \in \Re$ satisfy

$$\frac{\partial^2 f}{\partial x^2} = f'' \quad , \quad \frac{\partial^2 f}{\partial y^2} = i^2 f'' = - f'' \quad , \quad f'' = \frac{d^2 f}{dz^2}$$

$$\therefore \nabla^2 f = \nabla^2 u + i \nabla^2 v = 0 \quad \Rightarrow \quad \nabla^2 u = \nabla^2 v = 0$$

$\therefore$ using $f(z) = z^n$

$n = 1 \quad , \quad u = x \quad , \quad v = y$

$n = 2 \quad , \quad u = x^2 - y^2 \quad , \quad v = 2xy$

$n = 3 \quad , \quad u = x^3 - 3xy^2 \quad , \quad v = 3x^2 y - y^3$

$n = 4 \quad , \quad u = x^4 - 6x^2 y^2 + y^4 \quad , \quad v = 4x^3 y - 4xy^3$

Example 7- Torsion problem by BSM

Find $\hat{\phi}(x)$ solution of the following PDE

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -2 \qquad \text{in } \Omega : \{(x, y); -3 \le x \le 3; -2 \le y \le 2\}$$

$$\phi(x = -3) = \phi(x = 3) = 0$$

$$\phi(y = -2) = \phi(y = 2) = 0$$

equivalent to solve

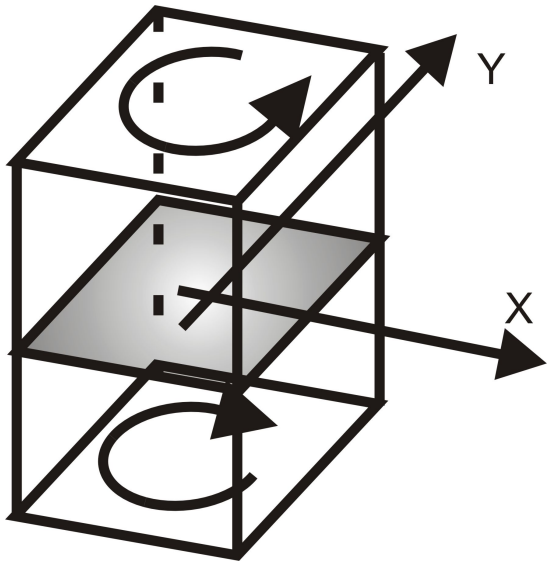
$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -2 \qquad \text{in } \Omega : \{(x, y); 0 \le x \le 3; 0 \le y \le 2\}$$

$$\phi(x = 3) = 0$$

$$\phi(y = 2) = 0$$

$$\left.\frac{\partial \phi}{\partial x}\right|_{x=0} = 0$$

$$\left.\frac{\partial \phi}{\partial y}\right|_{y=0} = 0$$



Boundary solution methods - Example

Changing variables

$$\phi = \theta - \frac{1}{2}\left(x^2 + y^2\right)$$

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = -2 \quad \Rightarrow \quad \frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0$$

Approximating θ

by symmetric functions respect to x and y

$$\theta \approx \hat{\theta} = a_1 \underbrace{1}_{N_1} + a_2 \underbrace{\left(x^2 - y^2\right)}_{N_2} + a_3 \underbrace{\left(x^4 - 6x^2 y^2 + y^4\right)}_{N_3}$$

$$\frac{\partial^2 \hat{\theta}}{\partial x^2} + \frac{\partial^2 \hat{\theta}}{\partial y^2} = 0 \quad , \quad \forall (x, y) \in \Omega$$

$$\int_{\Omega} \overline{W}_l \, R_{\Gamma} \, d\Gamma = 0$$

$$\Rightarrow \int_{\Omega} N_l \big|_{x=3} \left\{ \hat{\theta} \big|_{x=3} - \frac{1}{2}\left(9 + y^2\right) \right\} dy + \int_{\Omega} N_l \big|_{y=2} \left\{ \hat{\theta} \big|_{y=2} - \frac{1}{2}\left(x^2 + 4\right) \right\} dx = 0$$

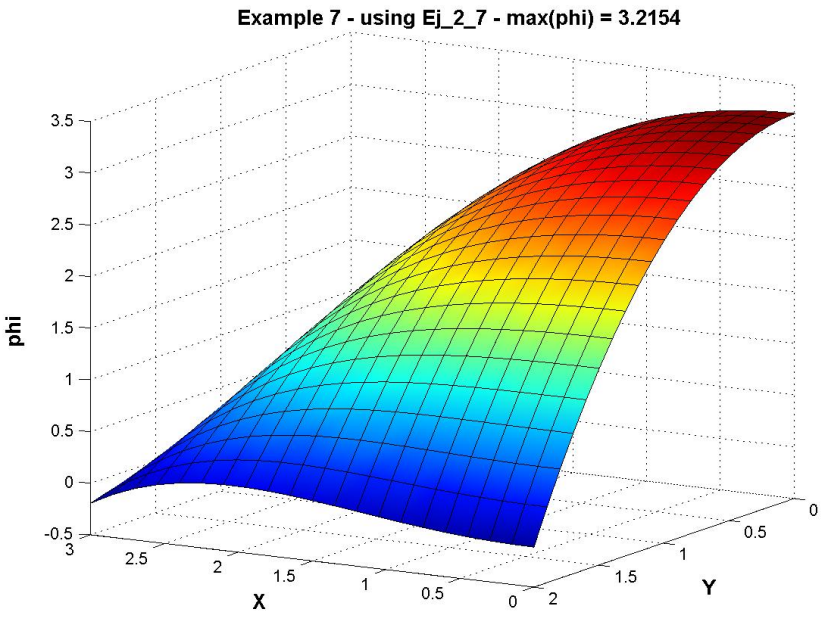
Boundary solution methods - Example

```
f = zeros(M,1);
K = zeros(M,M);
for ll=1:M
    f1 = gauss_integration('ffun_Ej_2_7_rhs_f1',1,c,d);
    f2 = gauss_integration('ffun_Ej_2_7_rhs_f2',1,a,b);
    f(ll) = -(f1+f2);
    for mm=1:M
        K1 = gauss_integration('ffun_Ej_2_7_lhs_f1',1,c,d);
        K2 = gauss_integration('ffun_Ej_2_7_lhs_f2',1,a,b);

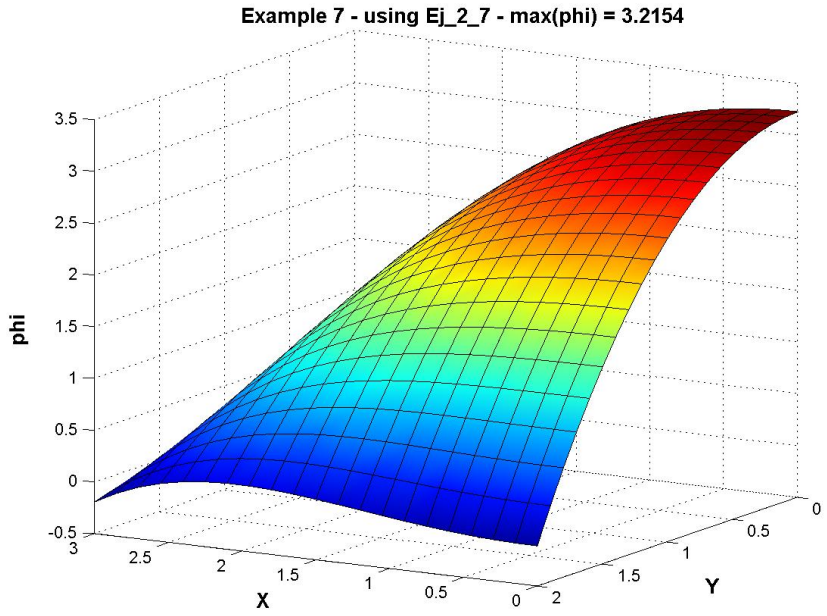
        K(ll,mm) = K1+K2;
    end
end
```

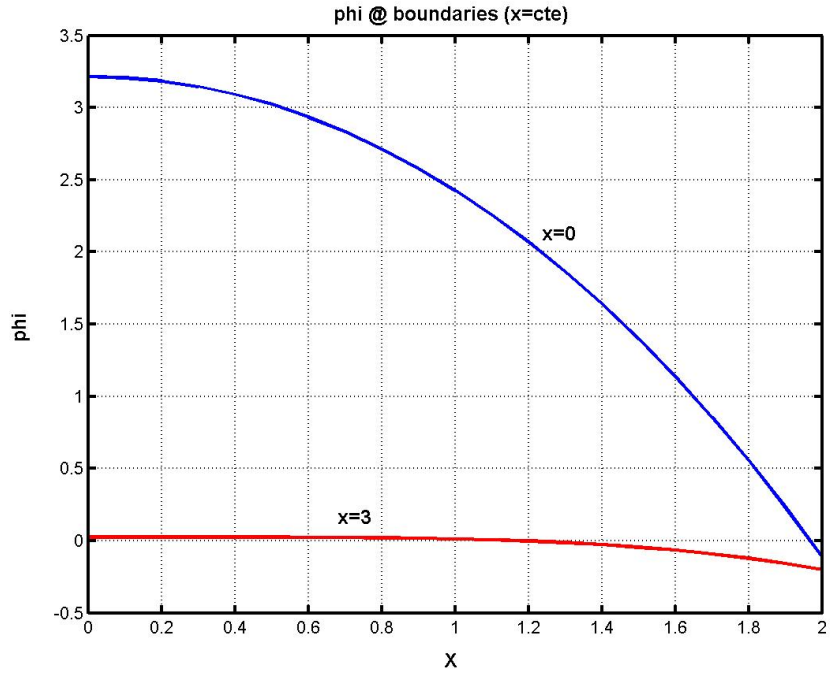
$$\int_{\Omega} N_l\big|_{x=3} \left\{ \hat{\theta}\big|_{x=3} - \frac{1}{2}\left(9 + y^2\right) \right\} dy +$$

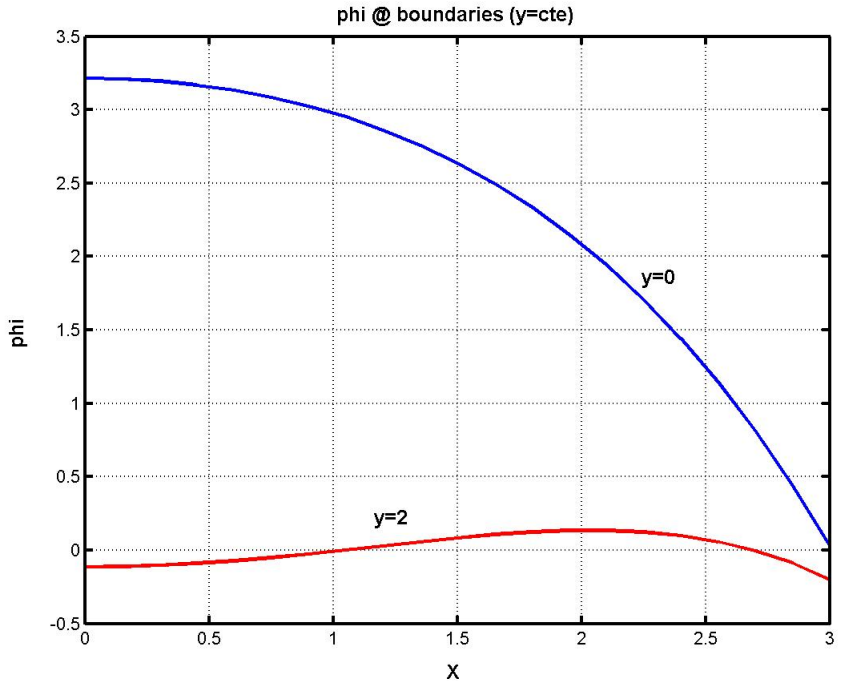
$$\int_{\Omega} N_l\big|_{y=2} \left\{ \hat{\theta}\big|_{y=2} - \frac{1}{2}\left(x^2 + 4\right) \right\} dx = 0$$



Boundary solution methods - Example







How the solution fits the right Dirichlet boundary values ?

Systems of differential equations

- Simultaneous resolution of PDEs coming from different applications
- 2D, axisymetric or 3D linear elasticity
- 1D beam equations
- 1D Gas dynamics
- Fluid mechanics in general
- Maxwell equations for electromagnetism
- Scalar 2nd order PDEs transformed to a system of 1st order PDE

Systems of differential equations

$$\underline{\phi}^T = (\phi_1, \phi_2, \phi_3, \ldots)$$

$$\begin{cases} A_1(\phi) = 0 \\ A_2(\phi) = 0 \\ \vdots \end{cases} \quad \text{in } \Omega \quad ; \quad A(\phi) = \begin{bmatrix} A_1(\phi) \\ A_2(\phi) \\ \vdots \end{bmatrix} = 0 \quad \text{in } \Omega$$

$$\begin{cases} B_1(\phi) = 0 \\ B_2(\phi) = 0 \\ \vdots \end{cases} \quad \text{in } \Gamma \quad ; \quad B(\phi) = \begin{bmatrix} B_1(\phi) \\ B_2(\phi) \\ \vdots \end{bmatrix} = 0 \quad \text{in } \Gamma$$

$$\phi_1(x) \cong \hat{\phi}_1(x) = \psi_1(x) + \sum_m a_{m,1} N_{m,1}(x)$$

$$\phi_2(x) \cong \hat{\phi}_2(x) = \psi_2(x) + \sum_m a_{m,2} N_{m,2}(x)$$

$$\vdots$$

$$\underline{\phi(x)} \cong \underline{\hat{\phi}(x)} = \underline{\psi(x)} + \sum_m \underline{a_m}\, \underline{\underline{N_m}}(x)$$

$$\underline{\psi(x)}^T = (\psi_1, \psi_2, \psi_3, \ldots) \quad ; \quad \underline{a_m}^T = (a_{m,1}, a_{m,2}, a_{m,3}, \ldots) \quad ; \quad \underline{\underline{N_m}} = \begin{bmatrix} N_{m,1} & & & 0 \\ & N_{m,2} & & \\ & & N_{m,3} & \\ 0 & & & \ddots \end{bmatrix}$$

Systems of differential equations

$$\int_{\Omega} W_{l,1}\, A_1\left(\hat{\phi}\right) d\Omega + \int_{\Gamma} \overline{W}_{l,1}\, B_1\left(\hat{\phi}\right) d\Gamma = 0$$

$$\int_{\Omega} W_{l,2}\, A_2\left(\hat{\phi}\right) d\Omega + \int_{\Gamma} \overline{W}_{l,2}\, B_2\left(\hat{\phi}\right) d\Gamma = 0$$

$$\vdots \qquad\qquad \vdots$$

$$\underline{\underline{W_l}} = \begin{bmatrix} W_{l,1} & & & 0 \\ & W_{l,2} & & \\ & & W_{l,3} & \\ 0 & & & \ddots \end{bmatrix} \quad ; \quad \underline{\underline{\overline{W_l}}} = \begin{bmatrix} \overline{W_{l,1}} & & & 0 \\ & \overline{W_{l,2}} & & \\ & & \overline{W_{l,3}} & \\ 0 & & & \ddots \end{bmatrix}$$

$$\int_{\Omega} \underline{W_l}\, \underline{A}\left(\underline{\hat{\phi}}\right) d\Omega + \int_{\Gamma} \underline{\overline{W}_l}\, \underline{B}\left(\underline{\hat{\phi}}\right) d\Gamma = 0$$

Example 8 : SoDE – 2nd order to 1st order

- Scalar heat equation (Poisson equation)
- In 1D
- 2nd order ODE (steady + 1D)
- Transformed to 1st order system of ODEs

$$\begin{cases} q + \kappa \dfrac{d\phi}{dx} = 0 \\ \dfrac{dq}{dx} - Q = 0 \end{cases} \quad ; \quad A(\phi) = \begin{bmatrix} q + \kappa \dfrac{d\phi}{dx} \\ \dfrac{dq}{dx} - Q \end{bmatrix} = 0 \quad \text{in } \Omega$$

$$\underline{\phi}^T = (q, \phi)$$

$$\mathsf{L}\,\underline{\phi} + \underline{p} = \begin{bmatrix} 1 & \kappa \dfrac{d}{dx} \\ \dfrac{d}{dx} & 0 \end{bmatrix} \underline{\phi} + \begin{bmatrix} 0 \\ -Q \end{bmatrix} = 0$$

Example 8 : SoDE – 2nd order to 1st order

- Scalar heat equation (Poisson equation)
- In 1D
- 2nd order ODE (steady + 1D)
- Transformed to 1st order system of ODEs

$$\Omega : \left\{ x; 0 \le x \le 1 \right\} \quad ; \quad k = 1 \quad ; \quad Q = \begin{cases} 1 & x \le \frac{1}{2} \\ 0 & x > \frac{1}{2} \end{cases}$$

$$\phi = 0 \quad @\, x = 0 \quad ; \quad q = 0 \quad @\, x = 1$$

$$\hat{q} = \psi_1 + \sum_m a_{m,1} N_{m,1}$$

$$\hat{\phi} = \psi_2 + \sum_m a_{m,2} N_{m,2}$$

$$\psi_1 = \psi_2 = 0 \quad ; \quad N_{m,1} = 0 \quad @\, x = 1 \quad ; \quad N_{m,2} = 0 \quad @\, x = 0$$

Example 8 : SoDE – 2nd order to 1st order

$$N_{m,1} = x^{m-1}(1-x) \quad ; \quad N_{m,2} = x^{m}$$

$$\int_0^1 \underline{W_l}\left(\mathsf{L}\,\underline{\hat{\phi}} + \underline{p}\right)dx = 0$$

$$K = \begin{bmatrix} K_{11} & K_{12} & \cdots & K_{1M} \\ K_{21} & K_{22} & \cdots & K_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ K_{M1} & K_{M2} & \cdots & K_{MM} \end{bmatrix} \quad ; \quad f = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_M \end{bmatrix}$$

$$K_{lm} = \int_0^1 \underline{W_l}\,\mathsf{L}\,\underline{N_m}\,dx \quad ; \quad f_l = -\int_0^1 \underline{W_l}\;\underline{p}\,dx$$

Galerkin type weighting $\underline{W_l} = \underline{N_l}$

$$K_{lm} = \begin{bmatrix} \int_0^1 N_{l,1} N_{m,1}\,dx & \int_0^1 N_{l,1}\dfrac{dN_{m,2}}{dx}\,dx \\ \int_0^1 N_{l,2}\dfrac{dN_{m,1}}{dx}\,dx & 0 \end{bmatrix}$$

Example 8 : SoDE – 2nd order to 1st order

```
f = zeros(M,ndf); K = zeros(M,M,ndf,ndf);

coef.ncol = ndf;
for ll=1:M
   coef.ncol = ndf;
   f(ll,1:ndf) = gauss_integration('ffun_Ej_2_8_rhs',0,1);
   for mm=1:M
      coef.ncol = ndf*ndf;
      vaux = gauss_integration('ffun_Ej_2_8_lhs',0,1);

      K(ll,mm,1:ndf,1:ndf) = reshape(vaux,ndf,ndf);
   end
end
```

Assembling

```
K2 = zeros(ndf*M,ndf*M); f2 = zeros(ndf*M,1);
for ll=1:M
   ll2 = (ll-1)*ndf+(1:ndf);
   f2(ll2) = reshape(f(ll,1:ndf),ndf,1);
   for mm=1:M
      mm2 = (mm-1)*ndf+(1:ndf);
      K2(ll2,mm2) = reshape(K(ll,mm,1:ndf,1:ndf),ndf,ndf);
   end
end

a = K2\f2;
```

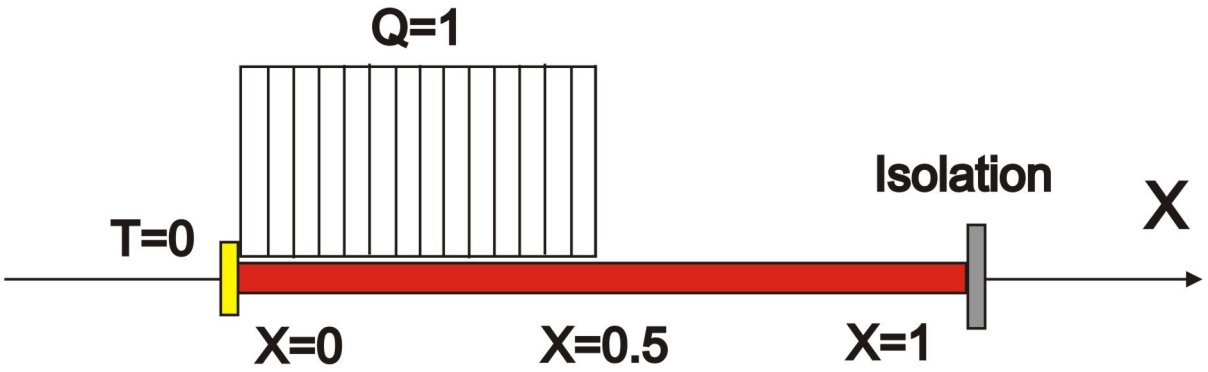
Reordering

Example 8 : SoDE – 2nd order to 1st order

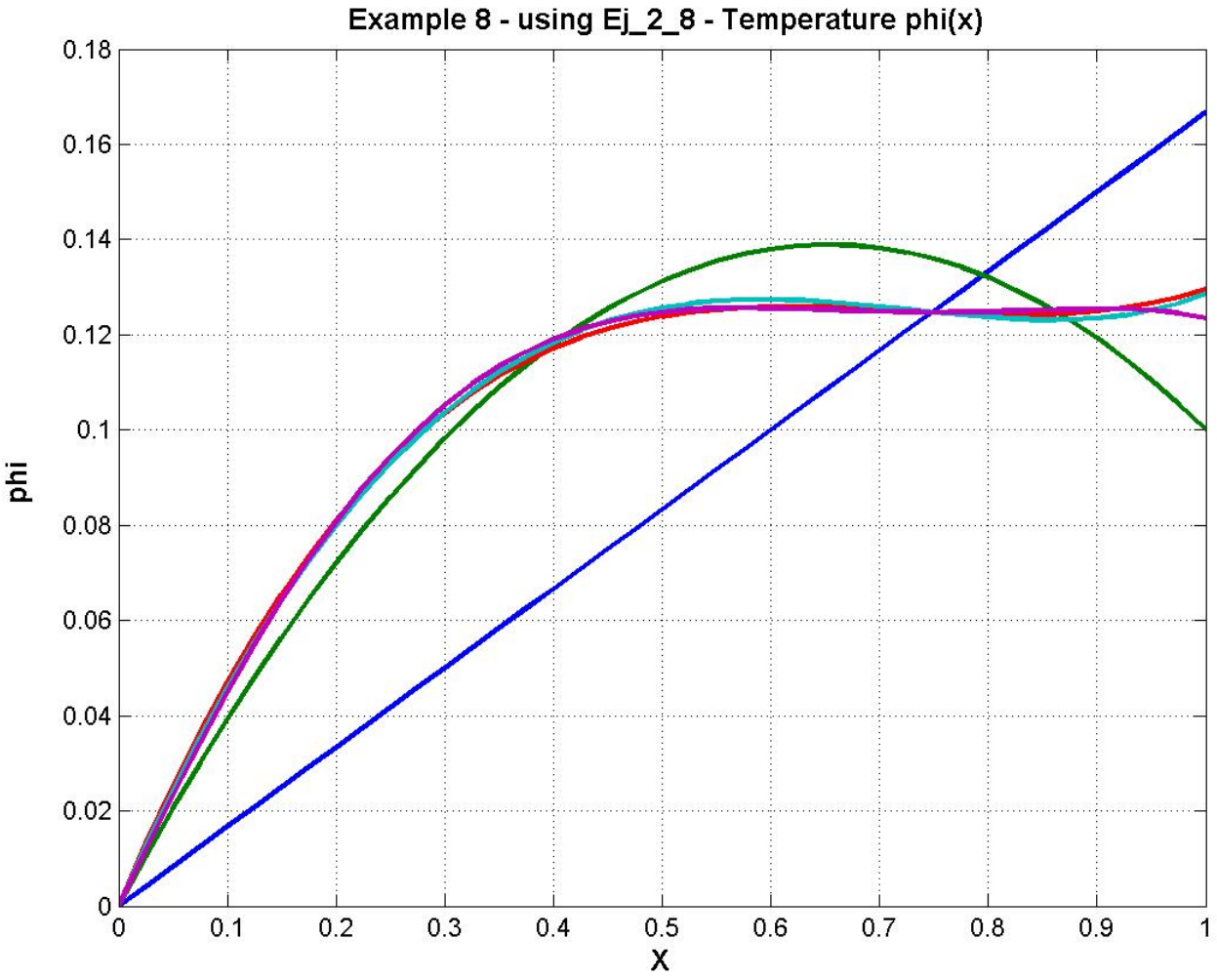
$$\frac{d}{dx}\left(\kappa\frac{d\phi}{dx}\right) - Q = 0$$

$$\Omega : \{x; 0 \le x \le 1\} \quad ; \quad \kappa = 1 \quad ; \quad Q = \begin{cases} 1 & x \le \frac{1}{2} \\ 0 & x > \frac{1}{2} \end{cases}$$

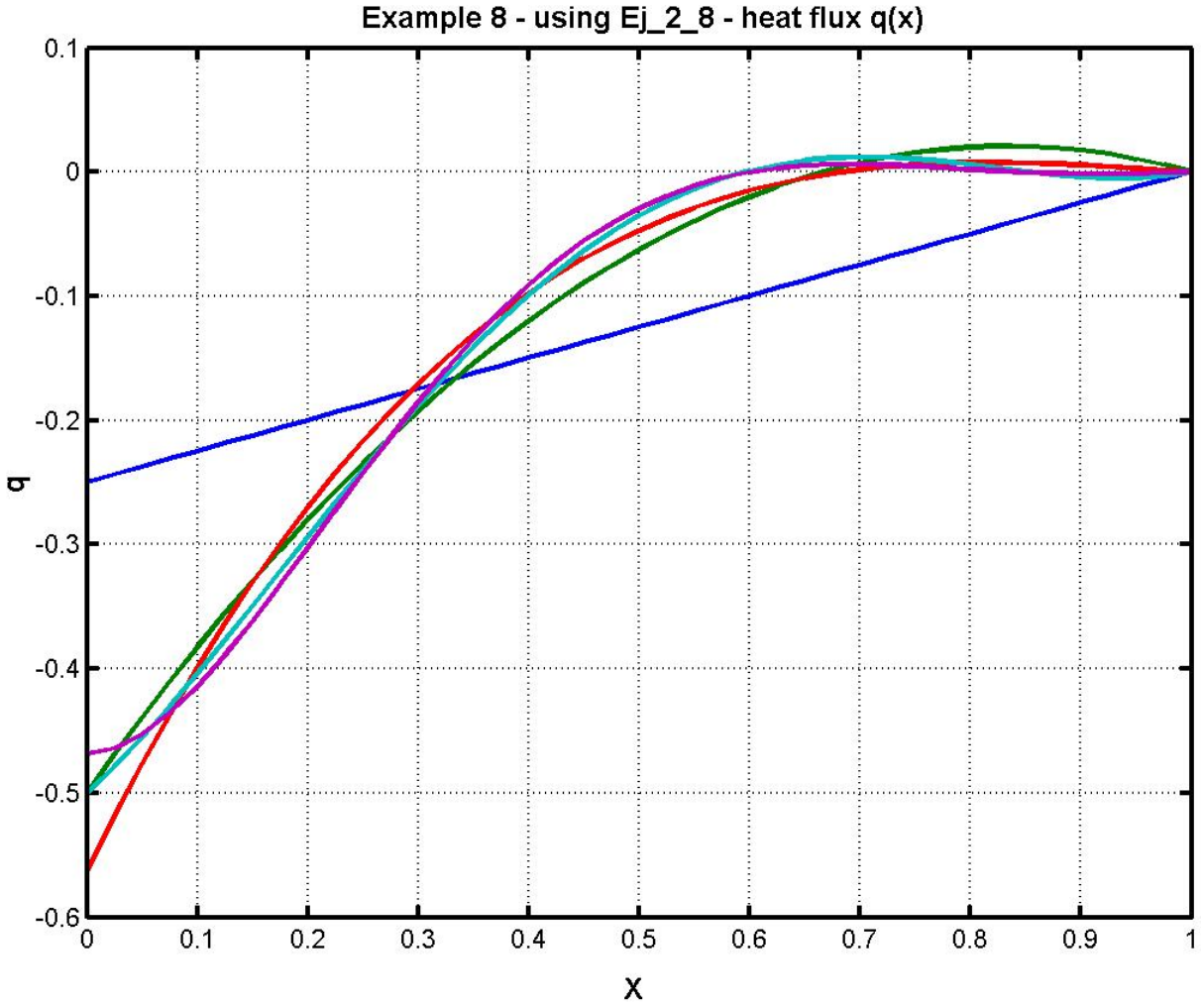
$$\phi = 0 \quad @\, x = 0 \quad ; \quad q = -\kappa\frac{d\phi}{dx} = 0 \quad @\, x = 1$$



Example 8 : SoDE – 2nd order to 1st order
Results – Temperature convergence with M



Example 8 : SoDE – 2nd order to 1st order
Results – heat flux convergence with M



Example 9 : SoDE – Linear elasticity in 2D

Plane stress in 2D elasticity

$$\varepsilon = \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_{xy} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial u}{\partial x} \\ \dfrac{\partial v}{\partial y} \\ \dfrac{\partial u}{\partial y} + \dfrac{\partial v}{\partial x} \end{bmatrix} = \mathsf{L}\,\underline{\phi} \quad ; \quad \mathsf{L} = \begin{bmatrix} \dfrac{\partial}{\partial x} & 0 \\ 0 & \dfrac{\partial}{\partial y} \\ \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix}$$

$$\sigma = \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_{xy} \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix} \varepsilon = \mathsf{D}\,\varepsilon$$

$$\underline{A}(\phi) = \begin{bmatrix} \dfrac{\partial \sigma_x}{\partial x} + \dfrac{\partial \sigma_{xy}}{\partial y} + X \\ \dfrac{\partial \sigma_{xy}}{\partial x} + \dfrac{\partial \sigma_y}{\partial y} + Y \end{bmatrix} = \mathsf{L}^{\mathsf{T}}\mathsf{D}\,\mathsf{L}\,\underline{\phi} + X = 0 \quad ; \quad \underline{X^T} = (X, Y)$$

$$\underline{B}(\phi) = \begin{bmatrix} \sigma_x n_x + \sigma_{xy} n_y - \hat{t}_x \\ \sigma_{xy} n_x + \sigma_y n_y - \hat{t}_y \end{bmatrix} = 0 \qquad \text{on } \Gamma_\sigma$$

$$\underline{B}(\phi) = \begin{bmatrix} u - \bar{u} \\ v - \bar{v} \end{bmatrix} = 0 \qquad \text{on } \Gamma_\phi$$

Example 9 : SoDE – Linear elasticity in 2D

Plane stress in

2D elasticity

$$\psi_1 = \bar{u} \quad ; \quad \psi_2 = \bar{v} \quad \text{on } \Gamma_\phi$$

$$\underline{\hat{\phi}} = \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix} = \underline{\psi} + \sum_m \underline{\underline{N_m}}\ \underline{a}_m \quad ; \underline{\psi}^T = (\psi_1, \psi_2)$$

$$\underline{\underline{W_l}} = \begin{bmatrix} W_{l,1} & \\ & W_{l,2} \end{bmatrix} \quad ; \quad \underline{\underline{\overline{W_l}}} = \begin{bmatrix} \overline{W_{l,1}} & \\ & \overline{W_{l,2}} \end{bmatrix}$$

$$\int_\Omega \left(\frac{\partial \sigma_x}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + X \right) W_{l,1}\, d\Omega + \int_{\Gamma_\sigma} (\sigma_x n_x + \sigma_{xy} n_y - \hat{t}_x)\, \overline{W_{l,1}}\, d\Gamma = 0$$

$$\int_\Omega \left(\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + Y \right) W_{l,2}\, d\Omega + \int_{\Gamma_\sigma} (\sigma_{xy} n_x + \sigma_y n_y - \hat{t}_y)\, \overline{W_{l,2}}\, d\Gamma = 0$$

$$\underline{\sigma} = \mathsf{D}\,\mathsf{L}\,\underline{\phi}$$

Example 9 : SoDE – Linear elasticity in 2D

Gauss - Green or integration by parts

$$- \int_{\Omega} \left(\hat{\sigma}_x \frac{\partial W_{l,1}}{\partial x} + \hat{\sigma}_{xy} \frac{\partial W_{l,1}}{\partial y} - W_{l,1} X \right) d\Omega + \int_{\Gamma_\sigma \cup \Gamma_\phi} \left(\hat{\sigma}_x n_x + \hat{\sigma}_{xy} n_y \right) W_{l,1} \, d\Gamma$$

$$+ \int_{\Gamma_\sigma} \left(\hat{\sigma}_x n_x + \hat{\sigma}_{xy} n_y - \hat{t}_x \right) \overline{W_{l,1}} \, d\Gamma = 0$$

$$- \int_{\Omega} \left(\hat{\sigma}_{xy} \frac{\partial W_{l,2}}{\partial x} + \hat{\sigma}_y \frac{\partial W_{l,2}}{\partial y} - W_{l,2} Y \right) d\Omega + \int_{\Gamma_\sigma \cup \Gamma_\phi} \left(\hat{\sigma}_{xy} n_x + \hat{\sigma}_y n_y \right) W_{l,2} \, d\Gamma$$

$$+ \int_{\Gamma_\sigma} \left(\hat{\sigma}_{xy} n_x + \hat{\sigma}_y n_y - \hat{t}_y \right) \overline{W_{l,2}} \, d\Gamma = 0$$

Example 9 : SoDE – Linear elasticity in 2D

Choosing the weighted functions as

$$W_{l,1} = W_{l,2} = 0 \qquad \text{on } \Gamma_\phi$$

$$\overline{W_{l,1}} = - W_{l,1}\Big|_{\Gamma_\sigma}$$

$$\overline{W_{l,2}} = - W_{l,2}\Big|_{\Gamma_\sigma}$$

$$- \int_\Omega \left(\hat{\sigma}_x \frac{\partial W_{l,1}}{\partial x} + \hat{\sigma}_{xy} \frac{\partial W_{l,1}}{\partial y} - W_{l,1} X \right) d\Omega + \int_{\Gamma_\sigma} \hat{t}_x \, W_{l,1} \, d\Gamma = 0$$

$$- \int_\Omega \left(\hat{\sigma}_{xy} \frac{\partial W_{l,2}}{\partial x} + \hat{\sigma}_y \frac{\partial W_{l,2}}{\partial y} - W_{l,2} Y \right) d\Omega + \int_{\Gamma_\sigma} \hat{t}_y \, W_{l,2} \, d\Gamma = 0$$

Example 9 : SoDE – Linear elasticity in 2D

Writing in a compact way

$$\mathsf{L}\,\underline{\hat{\phi}} = \mathsf{L}\,\underline{\psi} + \sum_m \mathsf{L}\,\underline{\underline{N}}_m\,\underline{a}_m$$

$$\int_\Omega \left(\mathsf{L}\,\underline{\underline{W}}_l\right)^T\ \underline{\hat{\sigma}}\,d\Omega - \int_\Omega \underline{\underline{W}}_l\,\underline{X}\ d\Omega - \int_{\Gamma_\sigma} \underline{\underline{W}}_l\,\underline{\hat{t}}\,d\Gamma = 0$$

$$\underline{\hat{t}} = \left(\hat{t}_x, \hat{t}_y\right)$$

expressing the stress in terms of displacements

$$\int_\Omega \left(\mathsf{L}\,\underline{\underline{W}}_l\right)^T \mathsf{D}\ \mathsf{L}\,\underline{\hat{\phi}}\,d\Omega = \int_\Omega \underline{\underline{W}}_l\,\underline{X}\ d\Omega + \int_{\Gamma_\sigma} \underline{\underline{W}}_l\,\underline{\hat{t}}\ d\Gamma$$

natural boundary condition

imposing $\underline{\hat{t}}$ is a natural boundary condition.

Example 9 : SoDE – Linear elasticity in 2D

using the approximation

$$\mathsf{L}\hat{\underline{\phi}} = \mathsf{L}\underline{\psi} + \sum_m \mathsf{L}\underline{\underline{N}}_m \, \underline{a}_m$$

$$\int_\Omega \left(\mathsf{L}\underline{\underline{W}}_l\right)^T \mathsf{D} \left(\mathsf{L}\underline{\psi} + \sum_m \mathsf{L}\underline{\underline{N}}_m \, \underline{a}_m \right) d\Omega = \int_\Omega \underline{\underline{W}}_l \, \underline{X} \; d\Omega + \int_{\Gamma_\sigma} \underline{\underline{W}}_l \, \underline{\hat{t}} \, d\Gamma$$

$$\sum_m \left(\int_\Omega \left(\mathsf{L}\underline{\underline{W}}_l\right)^T \mathsf{D} \; \mathsf{L}\underline{\underline{N}}_m \, d\Omega \right) \underline{a}_m = \int_\Omega \underline{\underline{W}}_l \, \underline{X} \; d\Omega + \int_{\Gamma_\sigma} \underline{\underline{W}}_l \, \underline{\hat{t}} \, d\Gamma -$$

$$- \int_\Omega \left(\mathsf{L}\underline{\underline{W}}_l\right)^T \mathsf{D} \; \mathsf{L}\underline{\psi} \, d\Omega$$

Example 9 : SoDE – Linear elasticity in 2D

linear system

$$\underline{\underline{K}}\,\underline{a} = \underline{f}$$

$$K_{lm} = \int_{\Omega} \left(\mathsf{L}\,\underline{\underline{W}}_l \right)^T \mathsf{D}\ \mathsf{L}\,\underline{\underline{N}}_m \, d\Omega$$

$$f_l = \int_{\Omega} \underline{\underline{W}}_l \,\underline{X}\ d\Omega + \int_{\Gamma_\sigma} \underline{\underline{W}}_l \,\underline{\hat{t}}\ d\Gamma - \int_{\Omega} \left(\mathsf{L}\,\underline{\underline{W}}_l \right)^T \mathsf{D}\ \mathsf{L}\,\underline{\psi}\, d\Omega$$

Example 9 : SoDE – Linear elasticity in 2D

Remark :

$$\int_{\Omega} \left(\mathsf{L}\, \underline{\underline{W_l}} \right)^T \hat{\sigma}\, d\Omega - \int_{\Omega} \underline{\underline{W_l}}\, \underline{X}\, d\Omega - \int_{\Gamma_\sigma} \underline{\underline{W_l}}\, \underline{\hat{t}}\, d\Gamma = 0$$

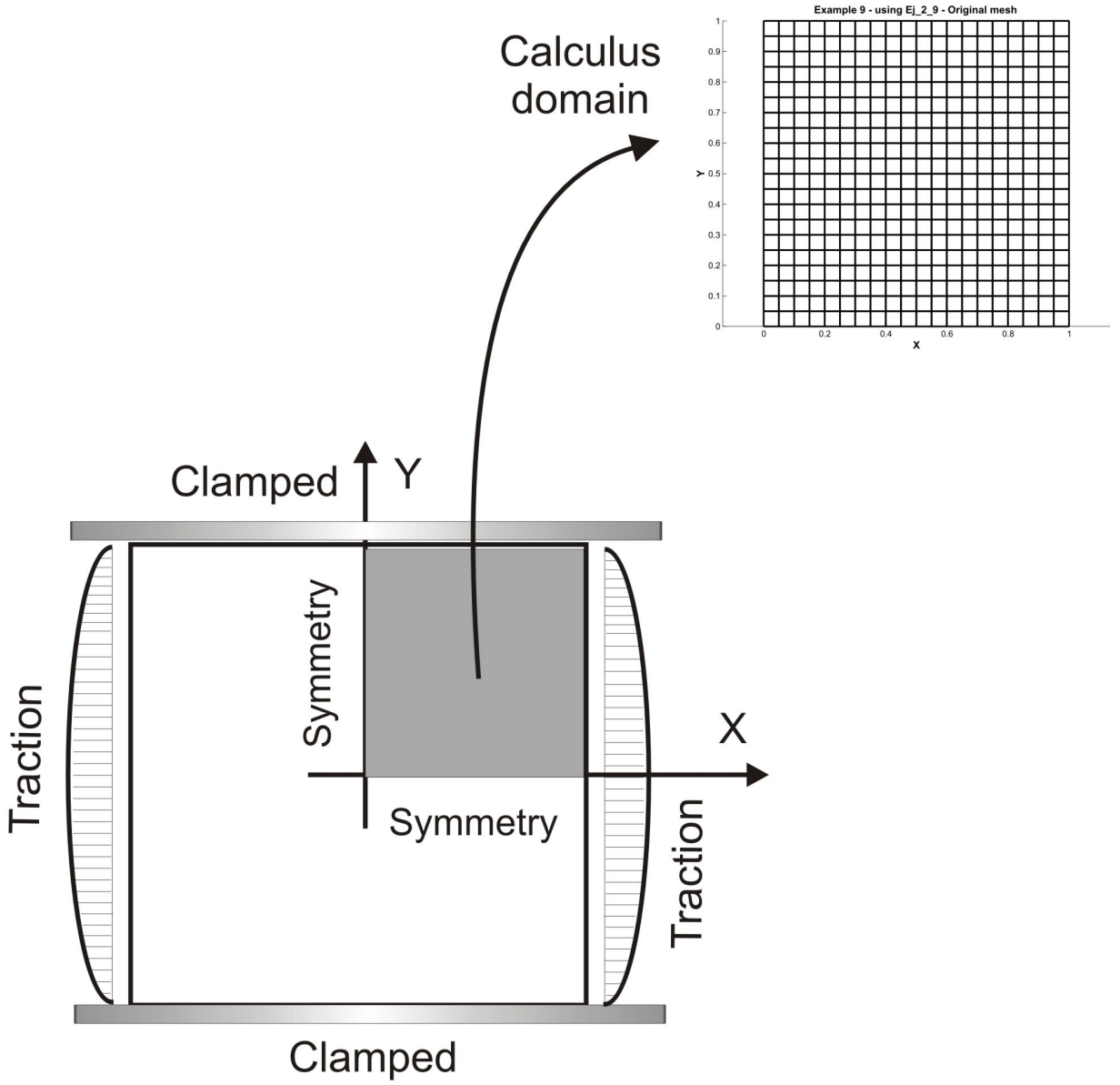
is equivalent to the principle of virtual work

$$\int_{\Omega} \left(\underline{\varepsilon}^* \right)^T \underline{\sigma}\, d\Omega = \int_{\Omega} \underline{\underline{\phi}}^{*T}\, \underline{X}\, d\Omega + \int_{\Gamma_\sigma} \underline{\underline{\phi}}^{*T}\, \underline{t}\, d\Gamma$$

using as virtual displacement $\underline{\underline{\phi}}^*$ the following definition

$$\underline{\underline{\phi}}^* = \underline{\underline{W_l}}\, \underline{\delta_l}^* \quad \text{with } \underline{\delta_l}^* \text{ arbitrary and} \quad \underline{\varepsilon}^* = \mathsf{L}\, \underline{\underline{\phi}}^*$$

Example 9 : Linear elasticity Problem definition (see Ej_2_9.m routine)



Example 9 : trial functions

```
Nxs =strvcat('(x)','(x.^3)','(x.*y.^2)');
dNxsdx =strvcat('1','(3*x.^2)','(y.^2)');
dNxsdy =strvcat('0','0','(2*y.*x)');

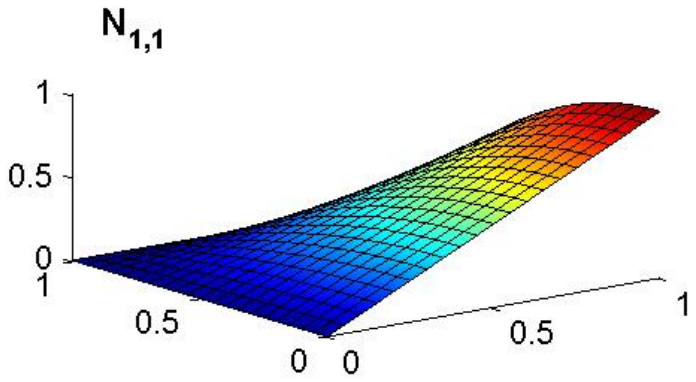
Nys =strvcat('(y)','(y.*x.^2)','(y.^3)');
dNysdx =strvcat('0','(2*y.*x)','0');
dNysdy =strvcat('1','(x.^2)','(3*y.^2)');

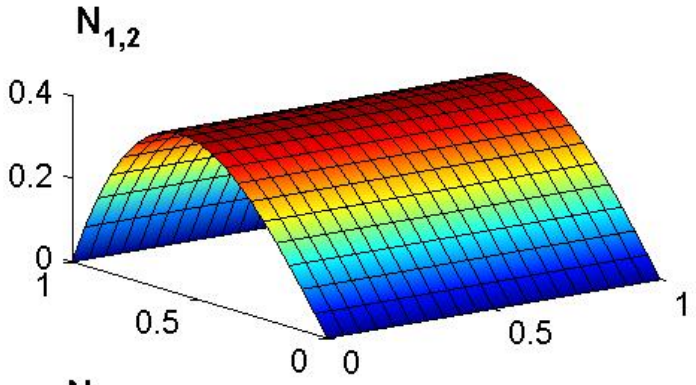
Nx = Nxs(mm,:);
dNxdx = dNxsdx(mm,:);
dNxdy = dNxsdy(mm,:);

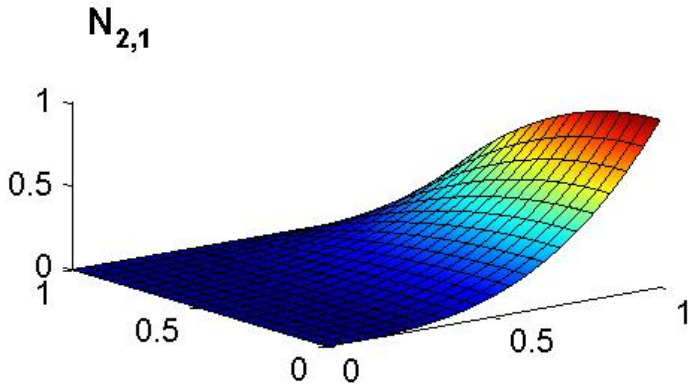
Ny = Nys(mm,:);
dNydx = dNysdx(mm,:);
dNydy = dNysdy(mm,:);

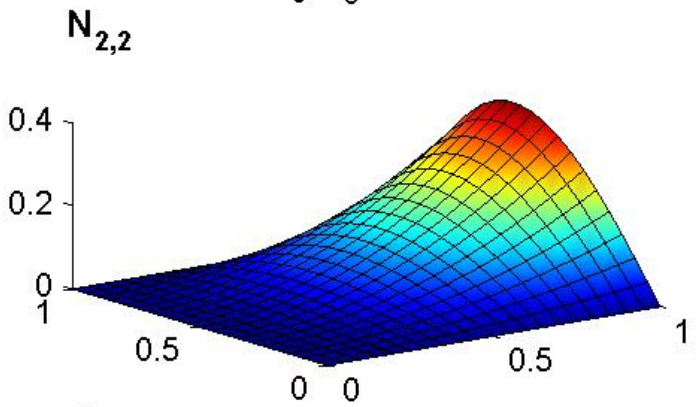
eval(['N1 = (1-y.^2).*' Nx ';']);
eval(['N2 = (1-y.^2).*' Ny ';']);
eval(['Lx_N1 = (1-y.^2).*' dNxdx ';']);
eval(['Ly_N1 = (1-y.^2).*' dNxdy '-' Nx '.*(2*y)' ';']);
eval(['Lx_N2 = (1-y.^2).*' dNydx ';']);
eval(['Ly_N2 = (1-y.^2).*' dNydy '-' Ny '.*(2*y)' ';']);
```

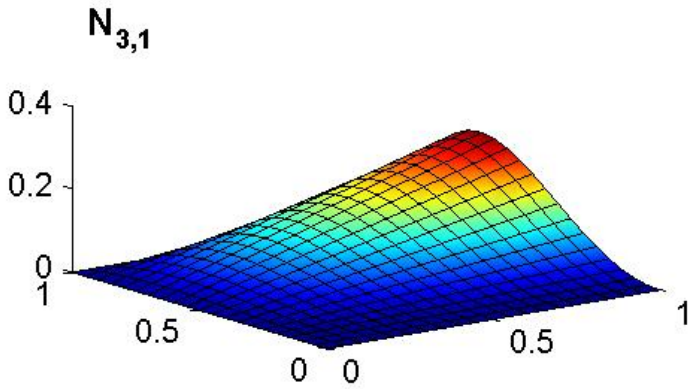
Example 9 : trial functions

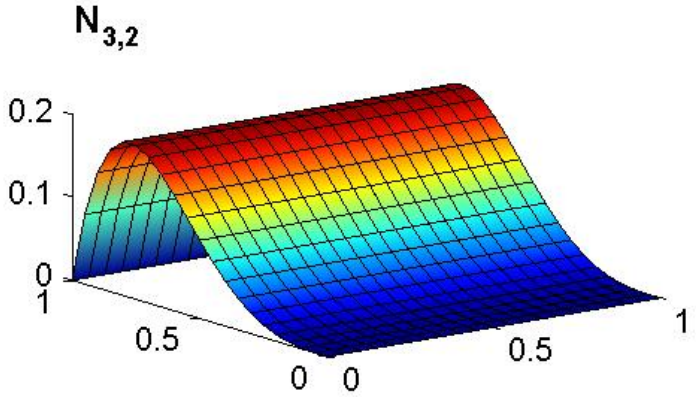




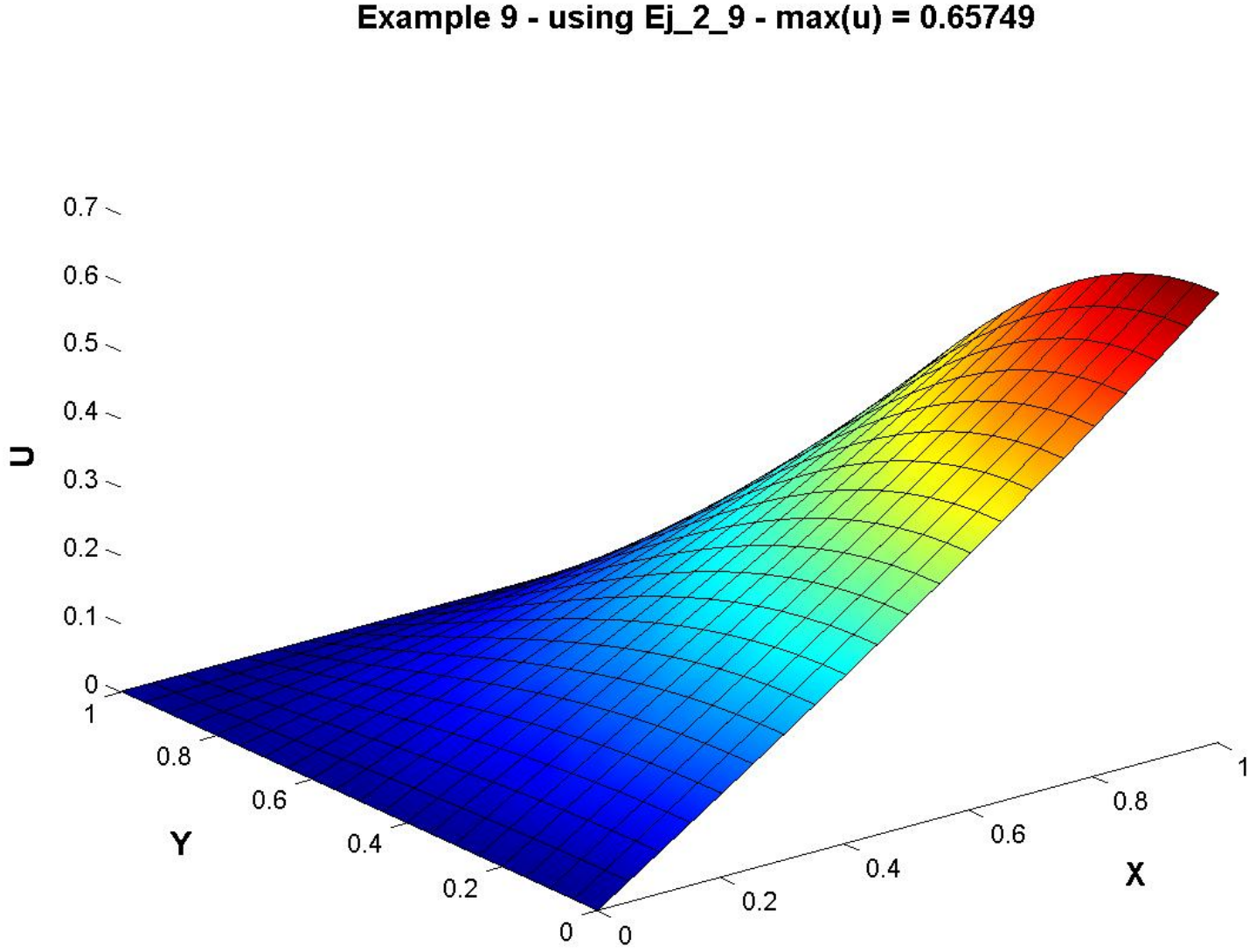




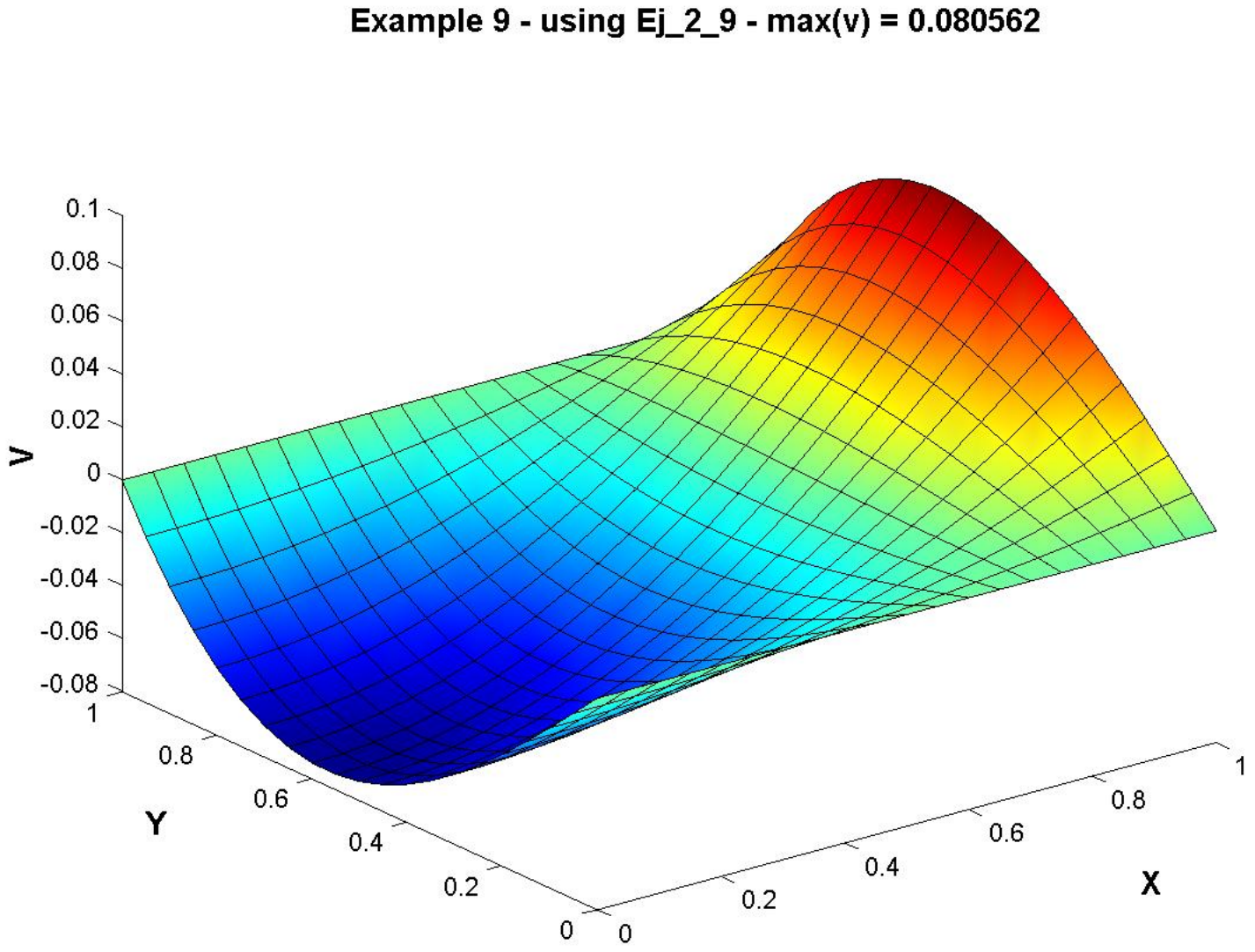




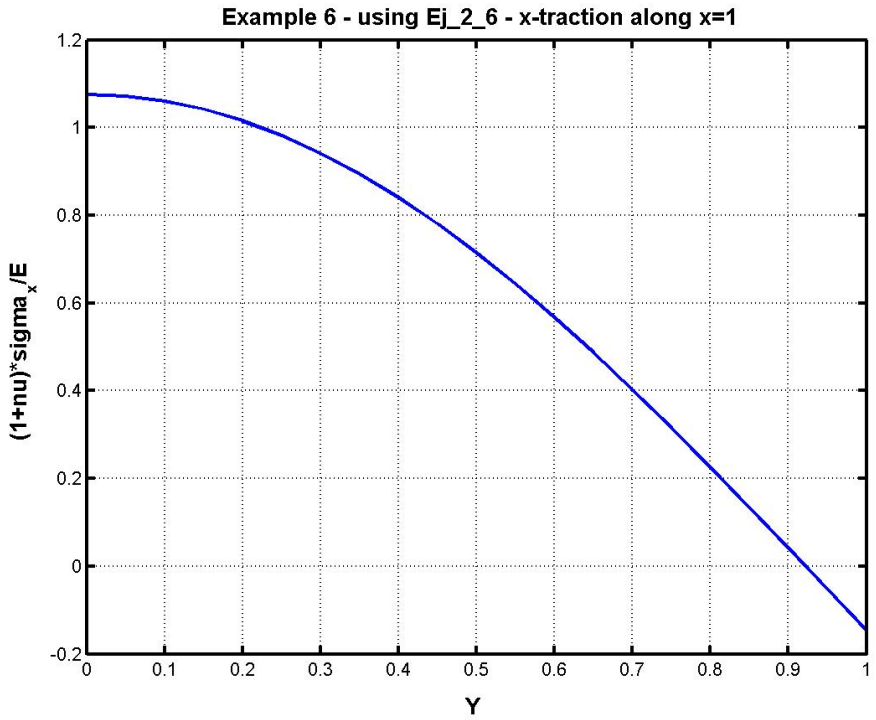
Example 9 : Linear elasticity – Results x-displacement

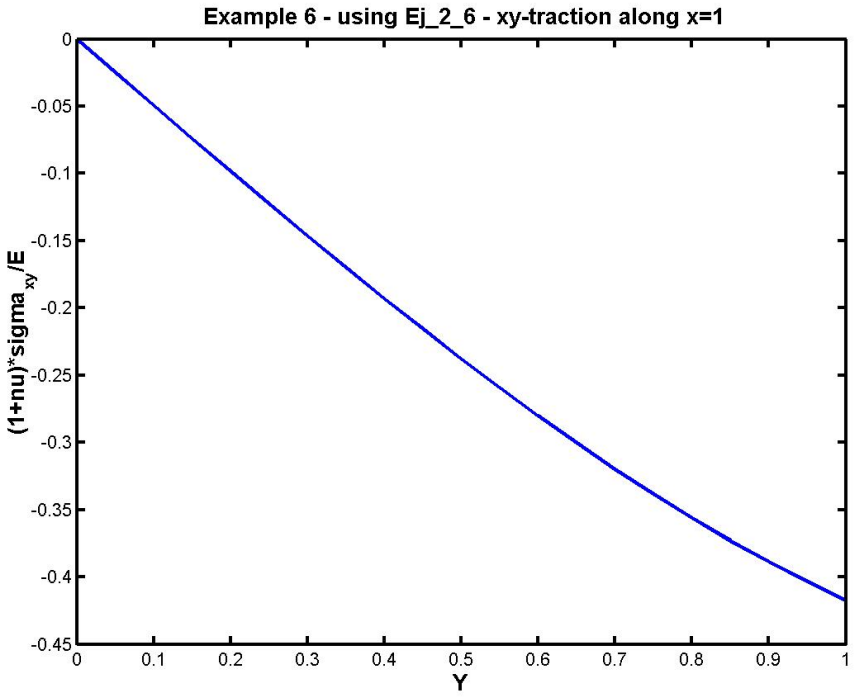


Example 9 : Linear elasticity – Results y-displacement

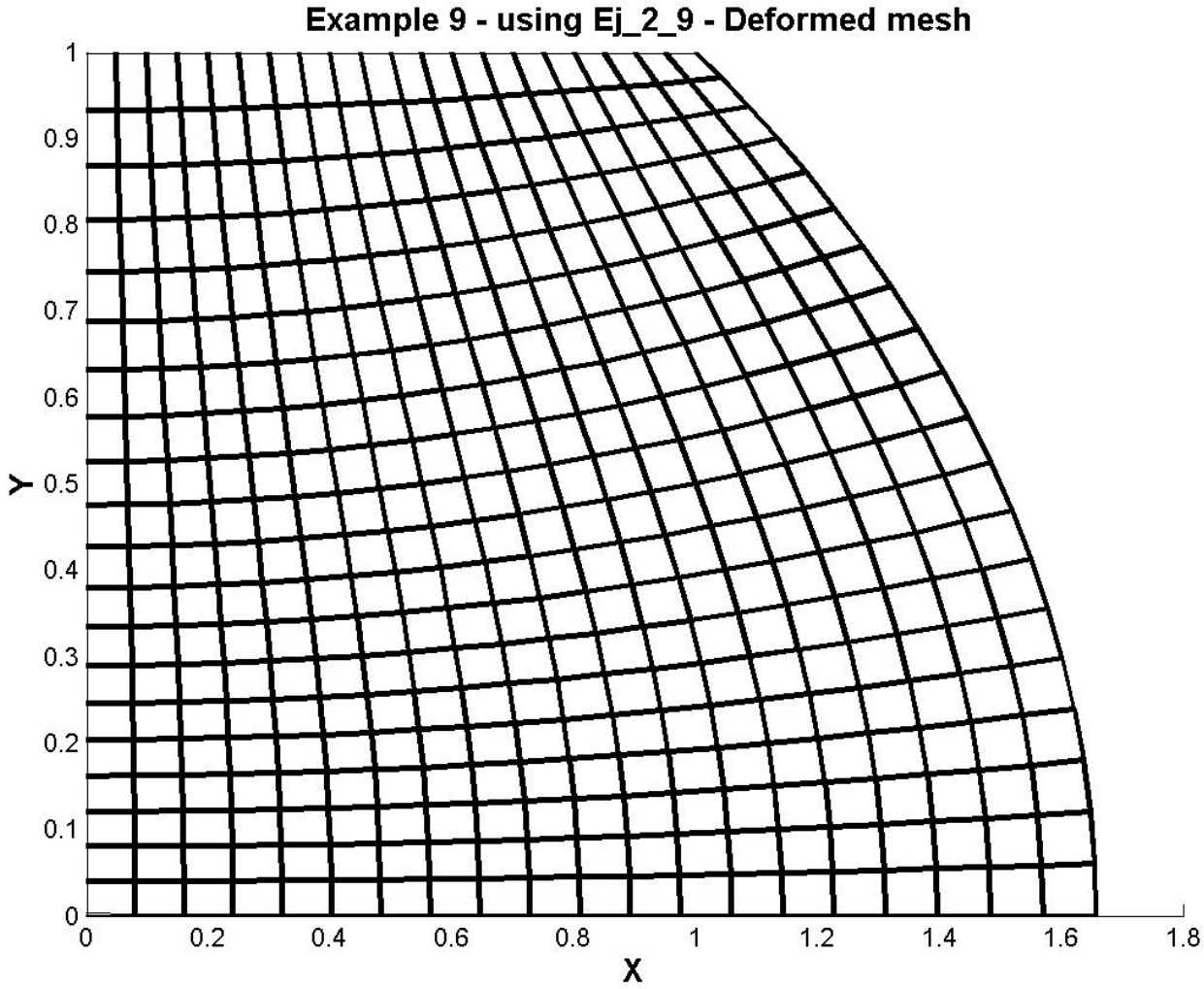


Example 9 : Linear elasticity – Results
Traction at boundary (x=1)





Example 9 : Linear elasticity – Results
Deformed mesh



Non linear problems

$$\text{Non linear problems} \Rightarrow \quad \mathrm{K(a)\,a} = \mathrm{f}$$

$$\text{Example}: \text{Find } \hat{\phi}(x) \text{ solution of the following PDE}$$

$$\frac{\partial}{\partial \mathrm{x}}\left(\kappa(\phi)\frac{\partial \phi}{\partial \mathrm{x}}\right)+\frac{\partial}{\partial \mathrm{y}}\left(\kappa(\phi)\frac{\partial \phi}{\partial \mathrm{y}}\right)+Q=0 \qquad \text{in } \Omega:\left\{\mathrm{x};\, x\in\Re^{2}\right\}$$

$$\phi=\bar{\phi} \quad \text{on } \Gamma_{\phi} \qquad ; \qquad \kappa\frac{\partial \phi}{\partial \eta}=-\bar{q} \quad \text{on } \Gamma_{q}$$

$$\hat{\phi}=\psi+\sum_{m} a_{m}N_{m}$$

$$\psi=\bar{\phi} \quad \text{on } \Gamma_{\phi} \quad ; \quad N_{m}=0 \quad \text{on } \Gamma_{\phi}$$

$$\int_{\Omega} W_{l}\left(\frac{\partial}{\partial \mathrm{x}}\left(\kappa\left(\hat{\phi}\right)\frac{\partial \hat{\phi}}{\partial \mathrm{x}}\right)+\frac{\partial}{\partial \mathrm{y}}\left(\kappa\left(\hat{\phi}\right)\frac{\partial \hat{\phi}}{\partial \mathrm{y}}\right)+Q\right)d\Omega+\int_{\Gamma_{q}}\overline{W_{l}}\left(\kappa\left(\hat{\phi}\right)\frac{\partial \hat{\phi}}{\partial \eta}+\bar{q}\right)d\Gamma=0$$

Non linear problems

$$\int_{\Omega} W_l \left(\frac{\partial}{\partial x} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial x} \right) + \frac{\partial}{\partial y} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial y} \right) + Q \right) d\Omega + \int_{\Gamma_q} \overline{W_l} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial \eta} + \bar{q} \right) d\Gamma = 0$$

integration by parts (Gauss - Green)

$$\int_{\Omega} \left(\frac{\partial W_l}{\partial x} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial x} \right) + \frac{\partial W_l}{\partial y} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial y} \right) - W_l Q \right) d\Omega -$$

$$- \int_{\Gamma_q \cup \Gamma_\phi} W_l \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial \eta} \right) d\Gamma - \int_{\Gamma_q} \overline{W_l} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial \eta} + \bar{q} \right) d\Gamma = 0$$

$$\overline{W_l} = - W_l \big|_{\Gamma_q} \quad ; \quad W_l \big|_{\Gamma_\phi} = 0$$

$$\int_{\Omega} \left(\frac{\partial W_l}{\partial x} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial x} \right) + \frac{\partial W_l}{\partial y} \left(\kappa(\hat{\phi}) \frac{\partial \hat{\phi}}{\partial y} \right) - W_l Q \right) d\Omega + \int_{\Gamma_q} W_l \bar{q} \; d\Gamma = 0$$

Non linear problems

$$K_{lm}(\underline{a}^0) = \int_\Omega \left(\frac{\partial W_l}{\partial x} \left(\kappa(\hat{\phi}^0) \frac{\partial N_m}{\partial x} \right) + \frac{\partial W_l}{\partial y} \left(\kappa(\hat{\phi}^0) \frac{\partial N_m}{\partial y} \right) \right) d\Omega$$

$$f_l(\underline{a}^0) = \int_\Omega W_l Q \, d\Omega - \int_{\Gamma_q} W_l \bar{q} \, d\Gamma -$$

$$- \int_\Omega \left(\frac{\partial W_l}{\partial x} \left(\kappa(\hat{\phi}^0) \frac{\partial \psi}{\partial x} \right) + \frac{\partial W_l}{\partial y} \left(\kappa(\hat{\phi}^0) \frac{\partial \psi}{\partial y} \right) \right) d\Omega +$$

In general, the iterations follow as :

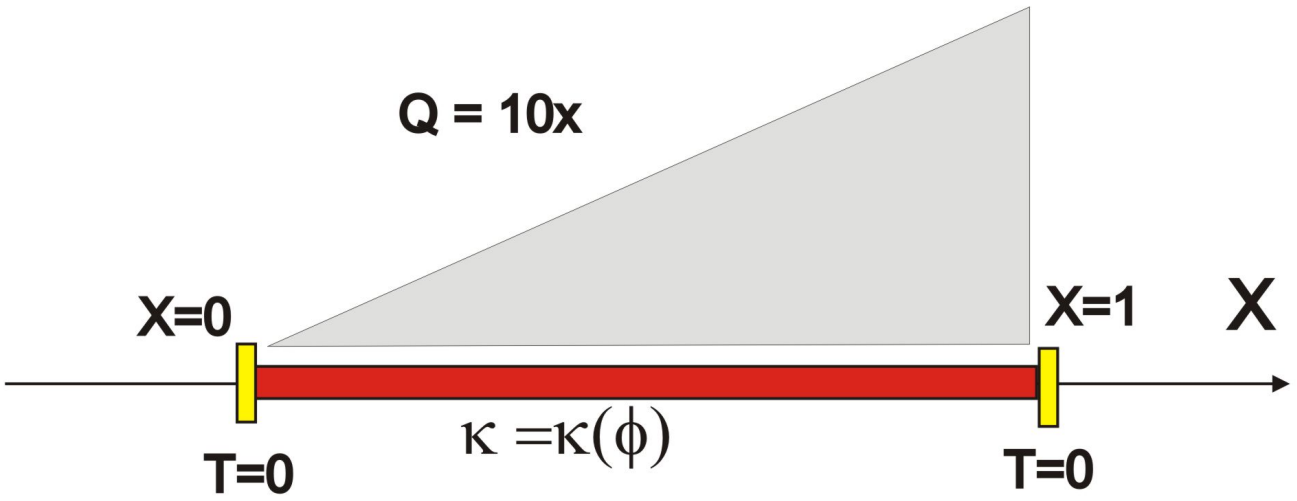
$$K(a^{n-1})a^n = f^{n-1}$$

Example 10 : Non linear heat equation

$$\frac{d}{dx}\left(\kappa\frac{d\phi}{dx}\right)+Q=0$$

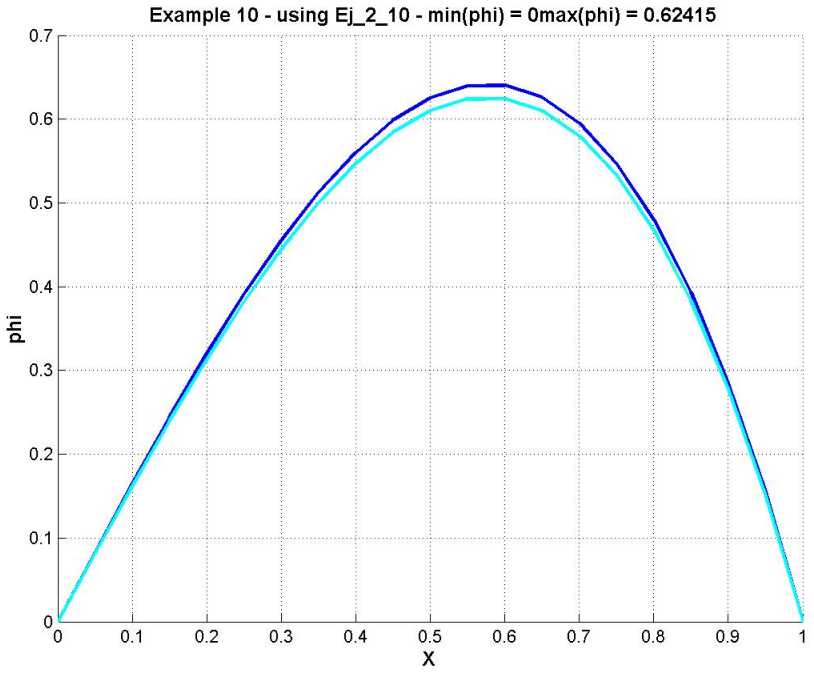
$$\Omega:\{x;0\le x\le 1\} \quad ; \quad \kappa=1+0.1\phi \quad ; \quad Q=10x$$

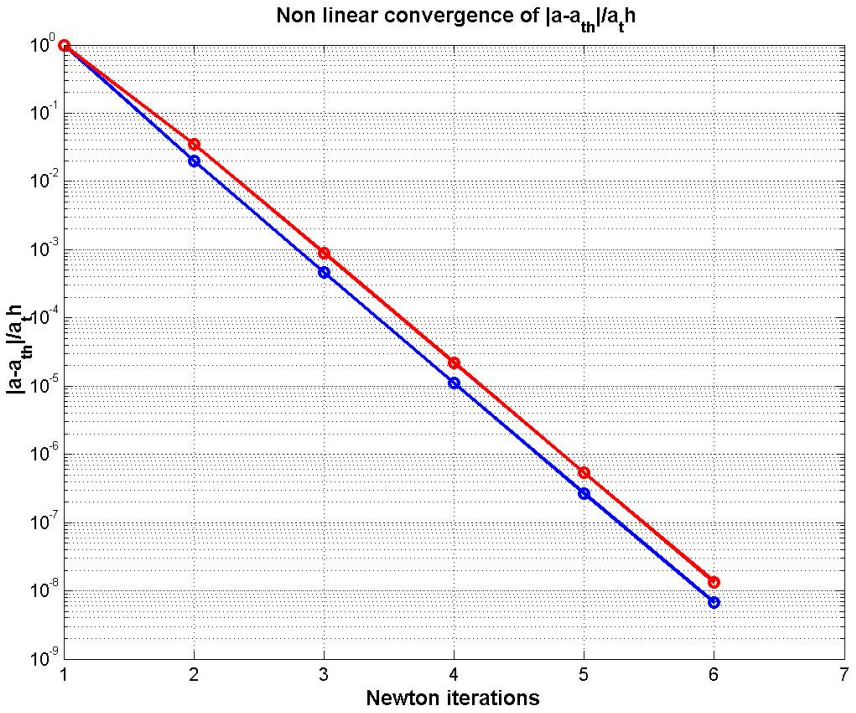
$$\phi=0 \quad @\,x=0 \quad ; \quad \phi=0 \quad @\,x=1$$



Example 10 : Non linear convergence of solution

M=2 – convergence of coefficients **a** with Galerkin using Ej_2_10.m routine





fin

